Local Digital Twins for Smart Sustainable Communities

LDT4SSC Methodology

Overview of the methodology and its workshops

Funded by the European Union

Cerema
CLIMAT & TERRITOIRES DE DEMAIN

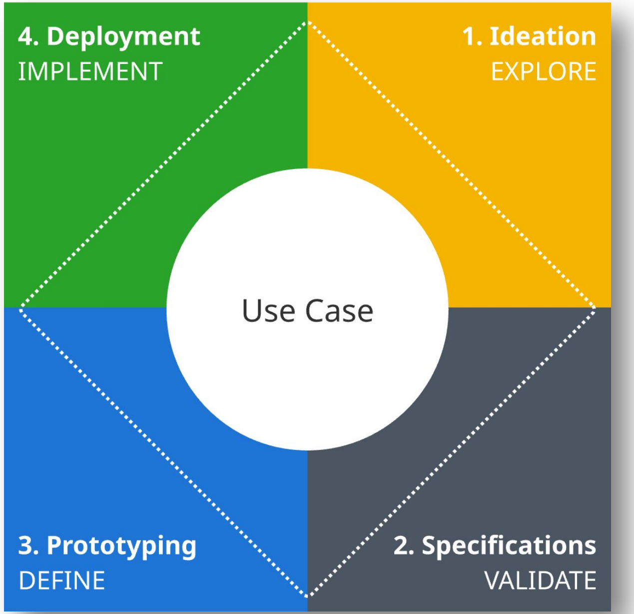
LDT4SSC Methodology

Introduction

The LDT4SSC Methodology is a **use-case-centered four-phase implementation framework** that guides LDT4SSC pilots from initial ideation through to full deployment of their projects.

This methodology provides a logical implementation framework for building a Local Digital Twin by structuring the pilot journey into four steps, each building on validated outputs from the previous one.

- 👉 **EXPLORE (Ideation)**: define user-centred use cases.
- 👉 **VALIDATE (Specifications)**: specifying the functionalities of the solution that meets the end users needs and setting up data governance.
- 👉 **DEFINE (Prototyping)**: testing the first solutions quickly and cost-effectively.
- 👉 **IMPLEMENT (Deployment)**: structuring an operational, scalable and sustainable project.



LDT4SSC Methodology

Based on the CAPACities Programme

The LDT4SSC methodology is based on the proceedings of the two editions of the French CAPACities Programme.

Launched by Cerema in 2022, these successive programmes supported 10 communities in building the skills and governance needed for a data-driven ecological transition. It brought together **local authorities, companies, and innovation partners** to co-develop smart territory projects based on real local challenges such as mobility, energy, resilience, and tourism.

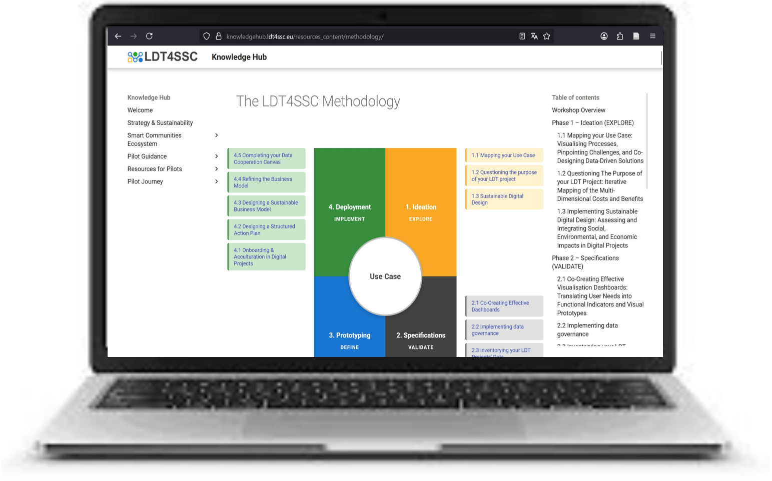
💡Through collaborative experimentation workshops, the programme provided **shared methods, tools, and data-governance frameworks** to accelerate the deployment of sustainable, interoperable, and locally managed digital solutions such as LDTs.

LDT4SSC Methodology

Methodology material

- A presentation page on the Knowledge Hub
- A global presentation deck of slides
- A document with instructions for carrying out each self-contained workshop

Visit the Methodology Knowledge Hub page



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Methodology workshops

The methodology integrates **15 practical and self-contained workshops** each tackling different challenges such as use case definition, data governance, interoperability, business modelling, and stakeholder onboarding.

- 📂 **Downloadable resources** are available for each step
- 🚨 Workshops don't cover everything there is to do in a given step but rather support its success

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What you will find in the workshop instructions documents

Summary: Title, Methodological step, Duration, Number of Facilitators, Equipment Needed and Expected Outputs

1. **Context and objectives**

   Methodology step and objectives of the workshop

2. **Workshop**

   2.1. **Workshop phases overview**

   Synthetic table of each phase in the workshop with its timing and description.

   2.2. **Workshop step by step guide**

   Description of the tasks to carry out and the facilitators' role for each workshop phase.

   Phase 0: Framing and Icebreaker

   Phase 1: …

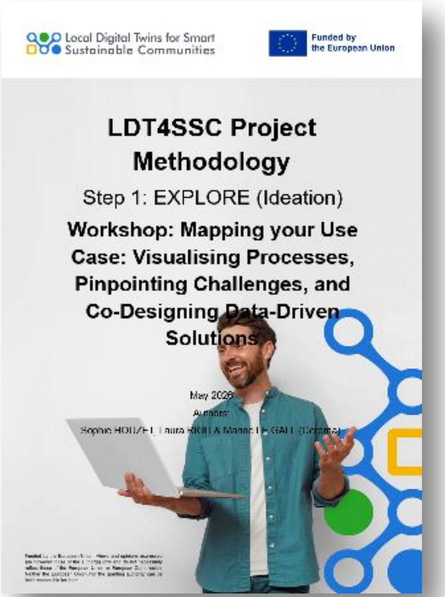
   Phase X: Synthesising and Defining Next Phases

   2.3. **Recommendations for Workshop facilitation**

   Help for facilitating the workshop and avoiding common pitfalls.

3. **Next Steps**

   Information about the next workshops to organise and methodology steps.



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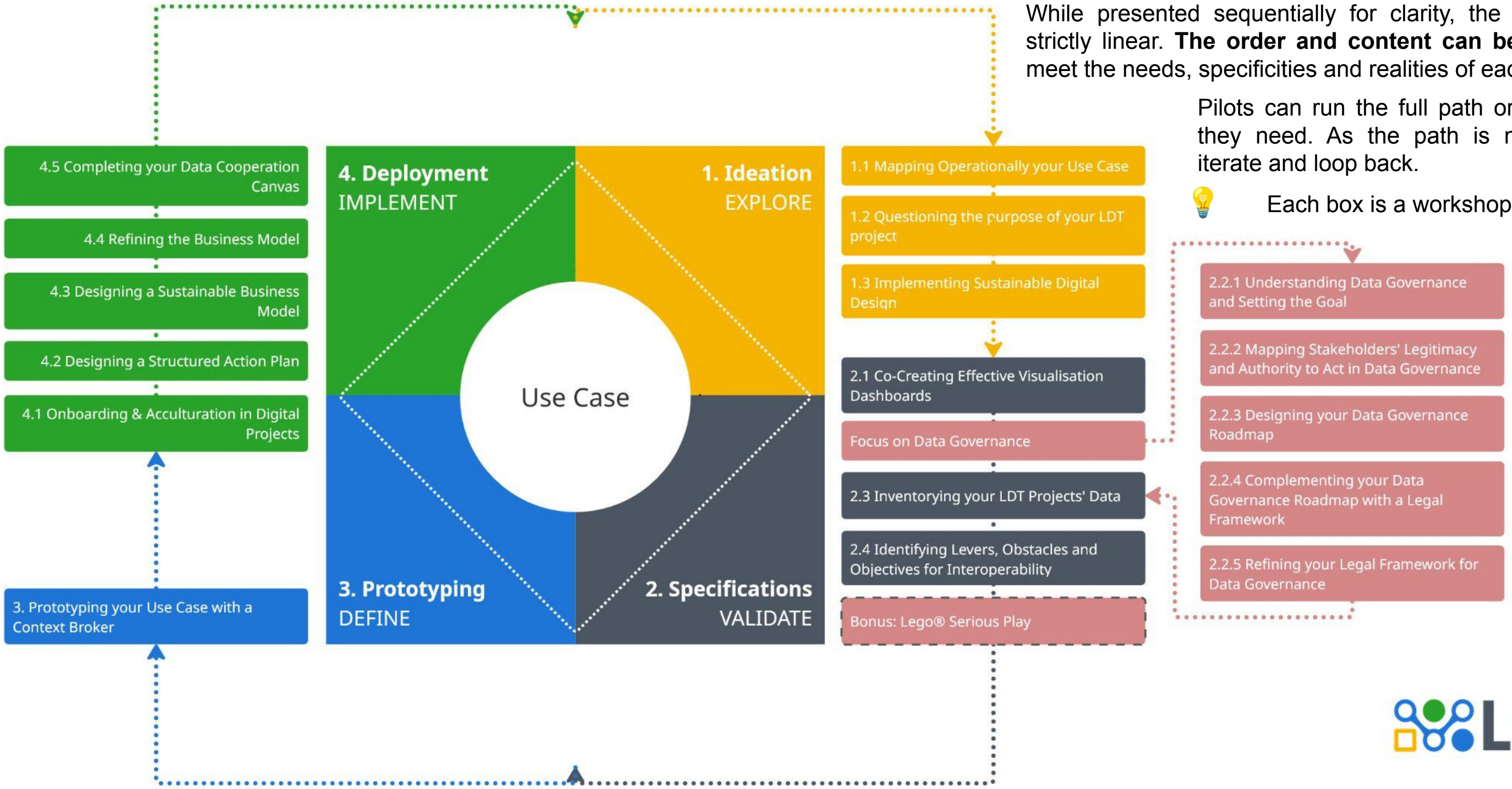
A use case-centered four-step implementation framework



While presented sequentially for clarity, the methodology is not strictly linear. **The order and content can be freely adapted** to meet the needs, specificities and realities of each project.

Pilots can run the full path or just the workshops they need. As the path is non-linear, they can iterate and loop back.

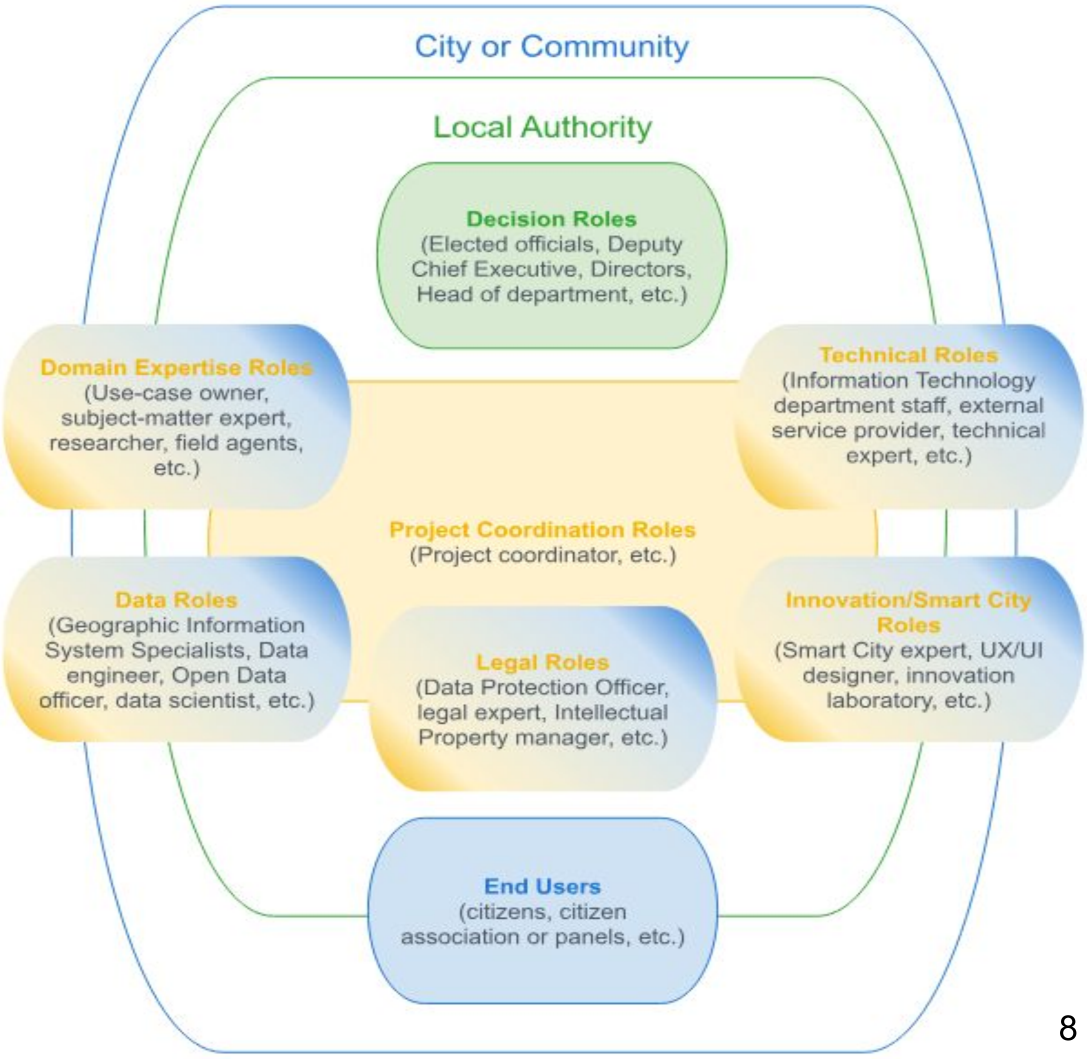
💡 Each box is a workshop !

Stakeholder Roles

Project-level Stakeholder Roles to involve in an LDT project

Roles to involve in each methodology steps are suggested in the material.

See deliverable 3.4 Non-Technical Resources for Pilots for more



1. EXPLORE

The Ideation step

LDT4SSC Methodology

1. EXPLORE

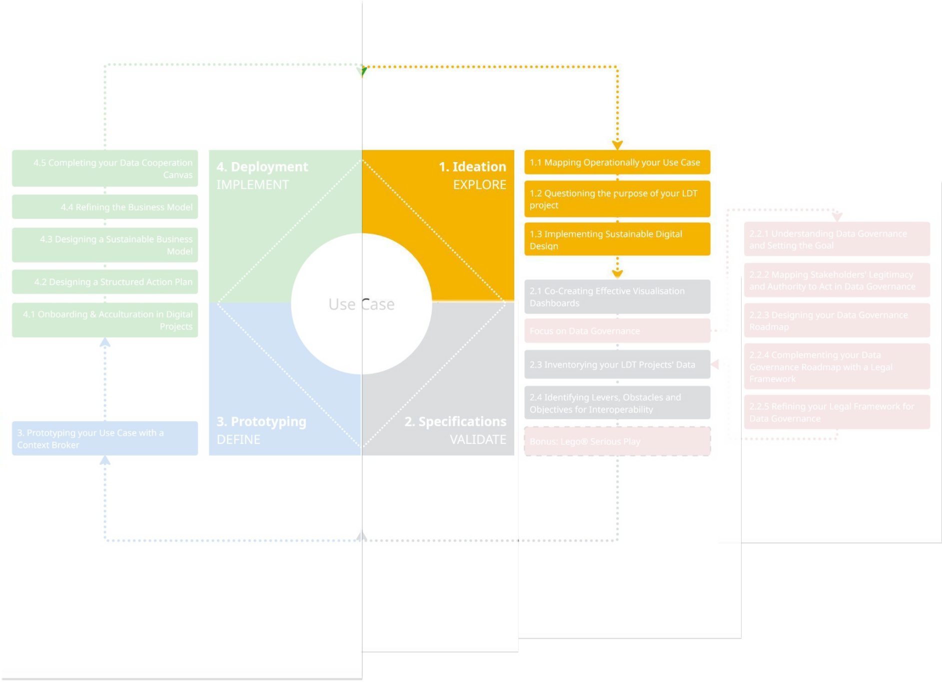
Introduction

Many digital projects begin with **predefined solutions** (dashboards, apps, algorithms) rather than **user needs**, risking misalignment, irrelevant data use, and stakeholder disengagement.

The **LDT4SSC Methodology** requires a **dedicated EXPLORE (Ideation) step** to:

- Clearly define the **use cases**.
- Identify **benefits and anticipate impacts**.
- Secure **stakeholder alignment**.

The Ideation step generally lasts from four to eight weeks, depending on the level of maturity and complexity of the project. It does **not necessarily follow a linear path, but is essential for the strategic grounding of the project**.
Some workshops are essential, in particular the **1.1 Mapping your Use Case**.



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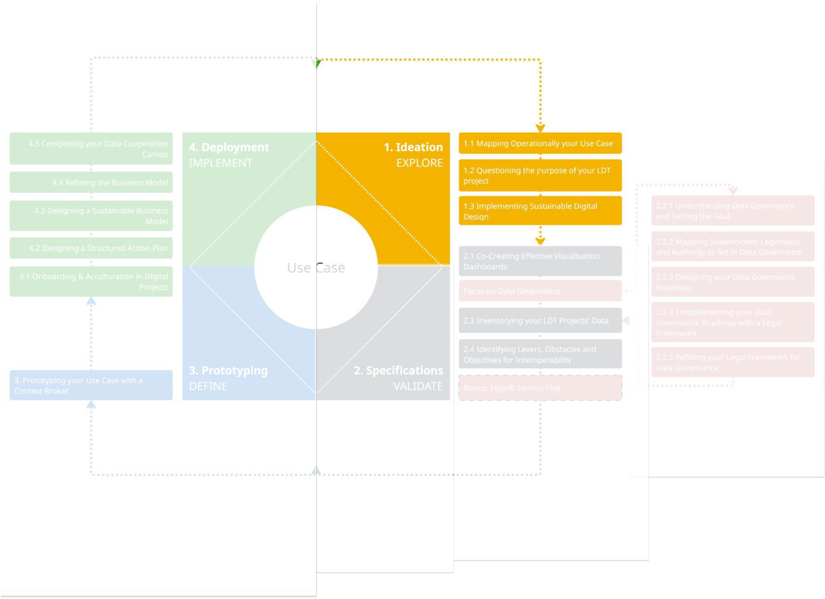
Prerequisite

1. EXPLORE

- A **broad idea of the project's goals** (e.g.,climate change adaptation, better traffic management, improved electricity-grid resilience).
- **Initial stakeholder identification and contact** (local authorities, experts, citizens, service providers).
- **Basic resource and data availability** for feasibility.
- **Project alignment with local public policies** (e.g., digital transformation, sustainability strategies, smart city objectives).



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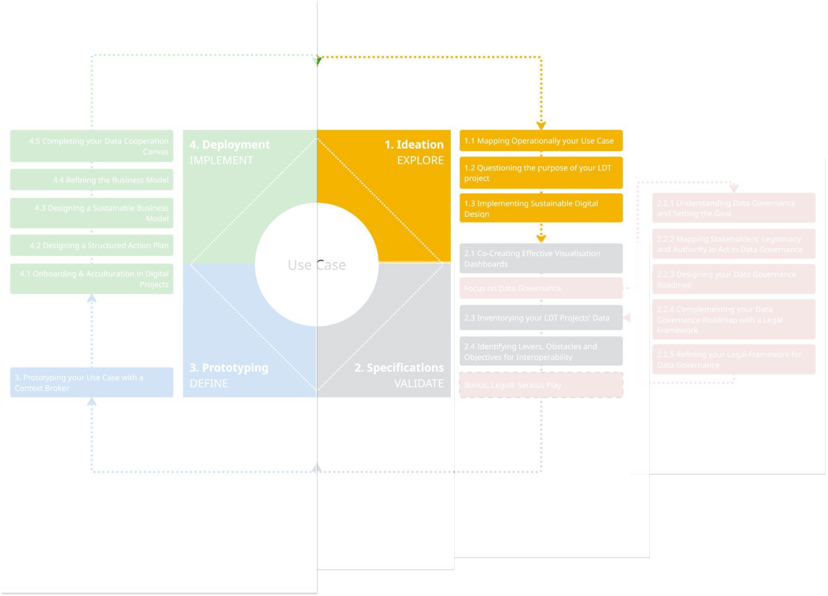
1. EXPLORE

High-level objectives

- 🎯 **Clearly define use cases** aligned with public policy and project goals.
- 🎯 **Engage stakeholders**, align them and understand their needs.
- 🎯 **Identify challenges and opportunities** for the Local Digital Twin (LDT) project.
- 🎯 **Assess benefits and impacts** (economic, social, environmental).
- 🎯 **Lay out the foundation** for the specification process and detailed planning in later steps.



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1. EXPLORE

Roles to involve

	Role	Rationale
Necessary	**Project Coordination Roles** (Project Coordinators, etc.)	They facilitate structured ideation, stakeholder alignment, and documentation of exploratory outputs.
	Domain Expertise Roles (Use-Case Owners, Experts, Researchers, Field Agents, etc.)	They define real-world challenges and opportunities, grounding ideation in practical, context-specific needs.
	Decision Roles (Elected Officials, Deputy Chief Executives, Directors, Heads of departments, etc.)	They ensure strategic alignment with policy priorities and secure high-level buy-in for project direction and resource allocation.
	End-Users or representatives of the **civil society** (Citizens, Citizens' Associations or Panels, etc.)	They ensure use cases reflect citizen priorities and societal impact, fostering inclusivity from the outset.
	Innovation / Smart City Roles (Smart City Experts, UX/UI Designers, Innovation Laboratories, etc.)	They drive user-centric, innovative design and ensure solutions align with smart city standards and emerging best practices.
Advised	**Technical Roles** (IIT Department Staff, External Service Provider, Technical Experts, etc.) **if available**	They provide early feasibility insights to avoid unrealistic assumptions and ground ideation in technical possibilities.
	Other Local Authorities facing similar issues	They offer peer-learning opportunities and best practices to accelerate problem-solving and innovation.

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1. EXPLORE

Workshops

#	Workshop	Duration
1.1	Mapping your Use Case	3h
1.2	Questioning The Purpose of your LDT Project	2h 40min × 3
1.3	Implementing Sustainable Digital Design	2h

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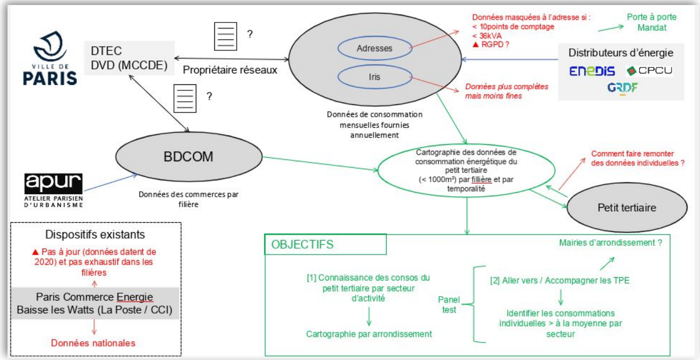
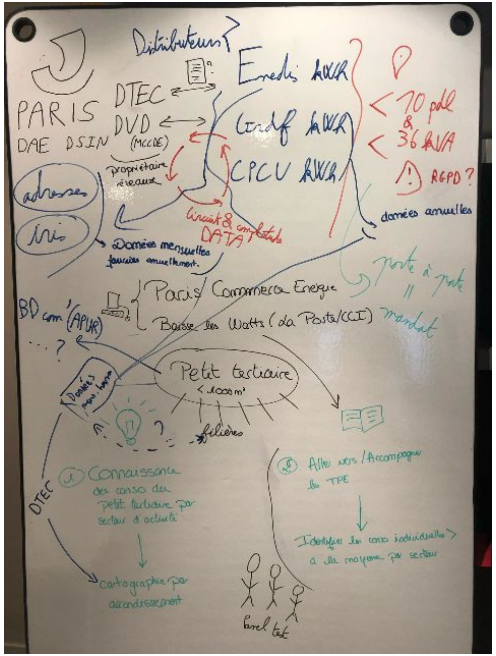
1. EXPLORE

1.1. Mapping your Use Case: Visualising Processes, Pinpointing Challenges, and Co-Designing Data-Driven Solutions

Workshop Objectives

- 🎯 **Visualise the end-to-end process** of your specific use case.
- 🎯 **Identify key challenges, bottlenecks, and opportunities** and address them for improvement.
- 🎯 **Co-design data-driven solutions** that align with stakeholder needs and project goals.





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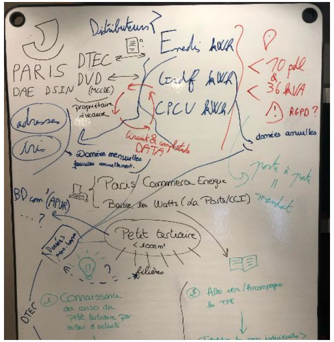
LDT4SSC Methodology

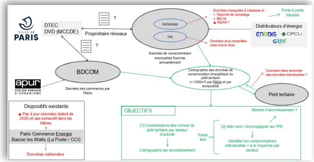
1. EXPLORE

1.1. Mapping your Use Case: Visualising Processes, Pinpointing Challenges, and Co-Designing Data-Driven Solutions

Workshop Overview

Title	Mapping Your Use Case: Visualising Processes, Pinpointing Challenges, and Co-Designing Data-Driven Solutions
Methodological step	Step 1: EXPLORE (Ideation)
Duration	3 hours
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • Pens and markers (black, blue, green, red).
Expected Outputs	• **Photo/scan of the produced diagram** (raw output from the session). • The **video recordings** of the summary presentations. • A **summary report** including: ○ Description of the **current use case**. ○ List of **identified challenges and gaps**. ○ List of **proposed data-driven solutions**. ○ **N**ext phases. • A **clean, digital version of the use case diagram** (finalized by the facilitator post-workshop for clarity and sharing).





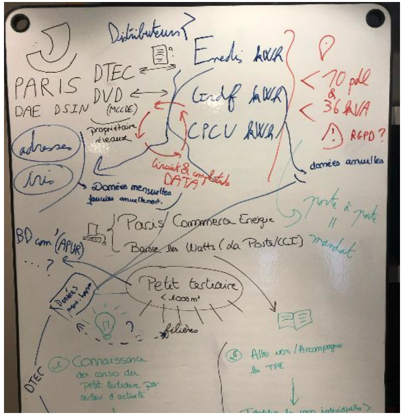
LDT4SSC Methodology

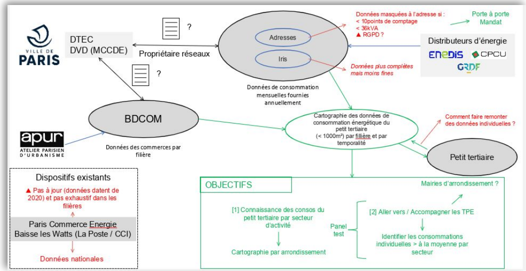
1. EXPLORE

1.1. Mapping your Use Case: Visualising Processes, Pinpointing Challenges, and Co-Designing Data-Driven Solutions

Workshop Outline

Time spent	***Phase***	***Description***
15 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
1 hour 30 mins	1.1. Mapping the current Use Case	Participants map the current use case, using colour coding to show data and data flows.
	1.2. Identifying the challenges	Participants identify and highlight challenges in the existing use case.
1 hour	2. Identifying opportunities and developing solutions	Participants identify opportunities and propose improvements to the use case.
15 mins	3. Summarising Group Insights	Each group presents the mapped use case, identified challenges and opportunities, and proposed solutions.





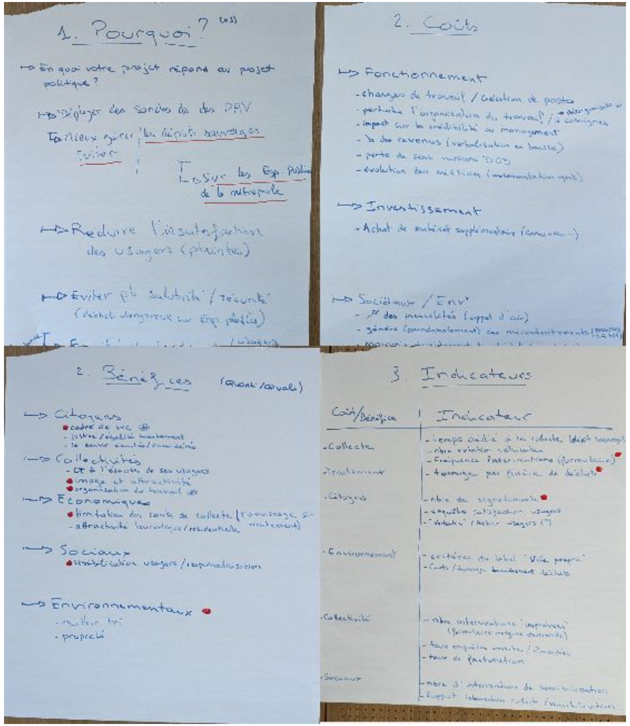
LDT4SSC Methodology

1. EXPLORE

1.2. Questioning the Purpose of your LDT Project: Iterative Mapping of the Multi-Dimensional Benefits

Workshop Objectives

- 🎯 Clarify the **strategic intent** and **public value** of the LDT project.
- 🎯 Identify and map **multidimensional benefits** (economic, social, environmental).



LDT4SSC

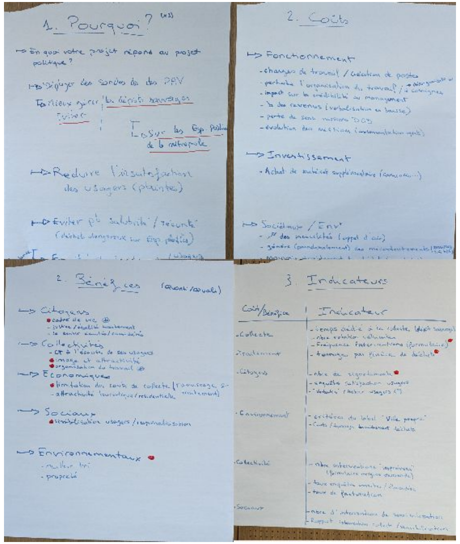
LDT4SSC Methodology

1. EXPLORE

1.2. Questioning the Purpose of your LDT Project: Iterative Mapping of the Multi-Dimensional Benefits

Workshop Overview

Title	Questioning The Purpose of your LDT Project: Iterative Mapping of the Multi-Dimensional Benefits
Methodological step	Step 1: EXPLORE (Ideation)
Duration	2 hours and 40 mins. Workshop to be carried out 3 times.
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • Pens and markers (blue, green, red).
Expected Outputs	• The filled-in Benefits Restitution Table. • A list of identified potential operational, strategic, social, and environmental benefits for the community • **A summary note** produced after the workshop, outlining key findings and next steps.



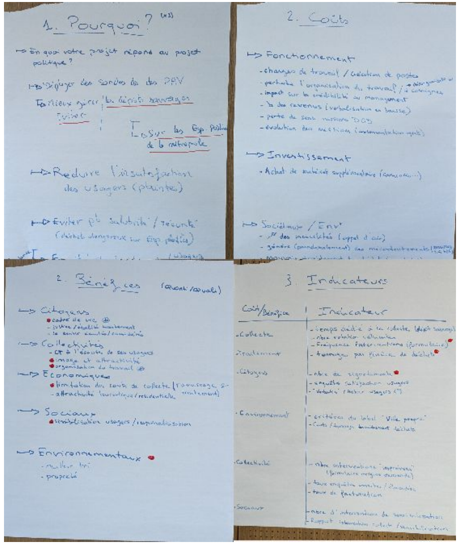
LDT4SSC Methodology

1. EXPLORE

1.2. Questioning the Purpose of your LDT Project: Iterative Mapping of the Multi-Dimensional Benefits

Workshop Outline

Time spent	***Phase***	***Description***
15 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
15 mins	1. Clearly defining the use cases purpose	The participants work on clarifying the project's purpose and clearly defining the use case.
45 mins	2. Identify the projects strategic benefits	Participants collaboratively identify strategic benefits and populate the benefits table.
20 mins	3. Refine the benefits	Participants project themselves five years into the future to reflect on the future benefits.
20 mins	4. Choosing indicators to measure the benefits	Participants refine project KPIs by proposing at least one relevant indicator for each benefit
30 mins	5.Presenting and Consolidating	Each sub-group presents their findings to the others.
15 mins	6. Synthesising and Defining Next Steps	Participants summarise the key findings of the workshop and draft follow-up actions.



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1. EXPLORE

1.3. Implementing Sustainable Digital Design: Assessing and Integrating Social, Environmental, and Economic Impacts in Digital Projects

Workshop Objectives

- 🎯 Perform preliminary **comprehensive assessment** of the LDTs impacts (social, environmental, economic).
- 🎯 Examine the digital project through the **lenses of sustainable and responsible digital design**.

In the CAPACities program, the workshop on sustainable digital design helped identify the positive and negative impacts in several key dimensions. For example, on the ecological front, measures such as server virtualization and query optimization have been identified to reduce the carbon footprint. However, negative impacts such as pollution and high initial costs have also been noted. Concrete actions, such as user awareness and resource optimization, have been proposed to mitigate these impacts.

	Conception	Usage	End of life	Positive impacts	Negative impacts	Actions?
Ecological / Energy dimension	Shared existing data			Frugality, energy efficiency		
Social dimension	Mobilization and inclusion of beneficiary users			Empower the actors of the small tertiary sector on this complex subject		
Economic/ Technical dimension	Fertilisation of existing services and common ones					
Ethical dimension						

The Sustainable Design Matrix

LDT4SSC Methodology

1. EXPLORE

1.3. Implementing Sustainable Digital Design: Assessing and Integrating Social, Environmental, and Economic Impacts in Digital Projects

Workshop Overview

Title	Implementing Sustainable Digital Design: Assessing and Integrating Social, Environmental, and Economic Impacts in Digital Projects
Methodological step	Step 1: EXPLORE (Ideation)
Duration	2 hours
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • Printed matrix or digital template; • Sticky notes, pens and markers.
Expected Outputs	• A filled in matrix of the LDTs' impacts and a shared understanding of them across sustainable and responsible digital design dimensions. • An initial action plan highlighting areas for improvement and opportunities for positive change. • **A summary note** produced after the workshop, outlining key findings and next steps.

	Conception	Usage	End of life	Positive impacts	Negative impacts	Actions ?
Ecological / Energy dimension	Shared existing data			Sobriety, energy efficiency		
Social dimension	Mobilization and inclusion of beneficiary users			Empower the actors of the small tertiary sector on this complex subject		
Economic/ Technical dimension	Fertilisation of existing services and common ones					
Ethical dimension						

The Sustainable Design Matrix

LDT4SSC Methodology

1. EXPLORE

1.3. Implementing Sustainable Digital Design: Assessing and Integrating Social, Environmental, and Economic Impacts in Digital Projects

Workshop Outline

Time spent	***Phase***	***Description***
15 mins	0.Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
1 hour	1.Exploring the Sustainable Design Matrix	Participants reflect on the LDTs' impacts using the sustainable and responsible digital design matrix.
30 mins	2. Consolidating Insights and Identifying Actions	Participants consolidate their insights and identify potential actions.
15 mins	3. Synthesising Outcomes and Defining Next Steps	The facilitator summarises the workshop key findings and defines follow-up actions.

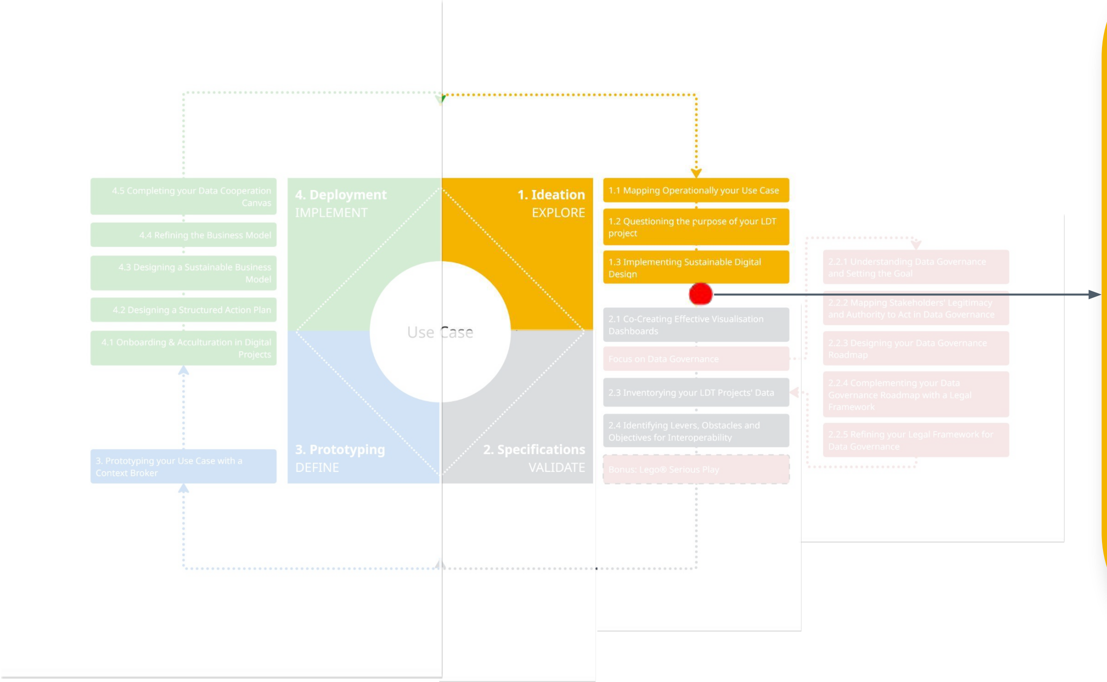
	Conception	Usage	End of life	Positive impacts	Negative impacts	Actions ?
Ecological / Energy dimension	Shared existing data			Sobriety, energy efficiency		
Social dimension	Mobilization and inclusion of beneficiary users			Empower the actors of the small tertiary sector on this complex subject		
Economic/ Technical dimension	Fertilisation of existing services and common ones					
Ethical dimension						

The Sustainable Design Matrix

LDT4SSC Methodology

1. EXPLORE

Achievements at the end of the EXPLORE step



At the end of this step of the methodology, the following elements have been completed or identified:

- **Clear project objectives** and alignment with public policy goals.
- **Shared understanding of use cases** reflecting stakeholder needs and challenges.
- **Sustainability and impact assessments** (economic, social, environmental).
- **Initial cost-benefit framework** to guide decision-making.
- **Collective understanding** of the project's purpose and strategic direction.

2. VALIDATE

The Specifications step

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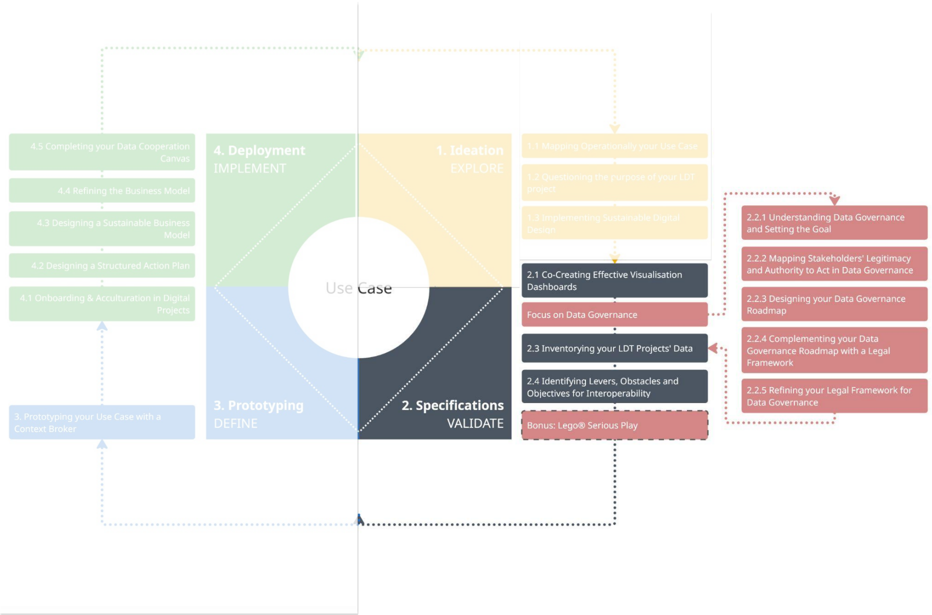
2. VALIDATE

Introduction



At the end of the explore (ideation) step, one should have a better idea of the need that should be met, the rationale behind the project, and with whom and with what data. However, it still remains **unclear how the project should be undertaken.**

The **VALIDATE (Specifications) step** provides the **link between the vision** that emerges from the Explore (Ideation) step and the **actual implementation** of the project. It involves translating a use case into **functional requirements**, taking into account the **real constraints of its environment** (data, architecture, users, security, etc.).

This step helps to **avoid misunderstandings between the business and technical teams**, and prepares a **solid basis for development**. It is also an opportunity to ensure that the **principles of interoperability** are integrated from the outset. It further support pilots in defining the following elements:

- The **personas**, i.e. the future users of the solution.
- Their **functionalities** in relation to users.
- The **indicators and dashboards** in relation to the personas.
- The desired **simulations.**
- The **data** and its governance.

NB: *Involving end-users is a key to success!*

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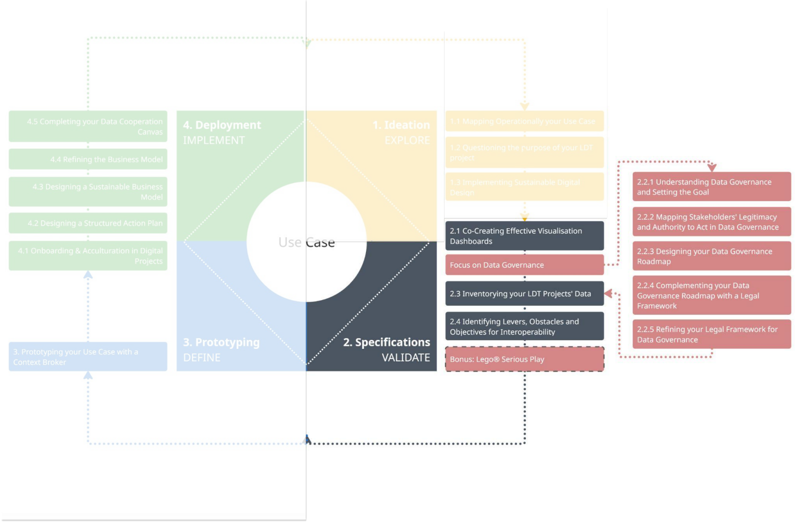
2. VALIDATE

Prerequisites

- 📝 **Completed the EXPLORE step**.
- 📝 **Access to technical, legal, and operational expertise** for feasibility assessments.
- 📝 **Stakeholder commitment and availability** for validation discussions.
- 📝 **High level buy-in** of the project from elected officials.
- 📝 **Clear criteria for evaluation** (e.g., impact metrics, compliance requirements, resource constraints).



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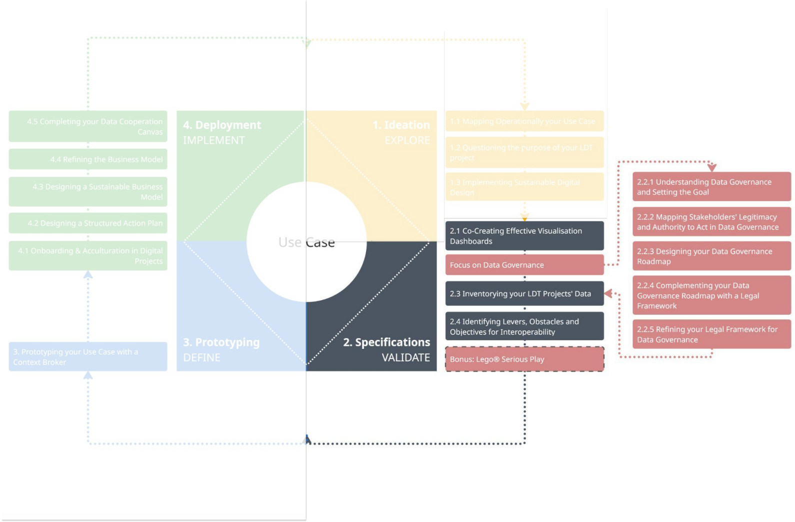
2. VALIDATE

High-level objectives

- **Refine and validate** use cases definitions, hypotheses, and assumptions from the EXPLORE step.
- **Assess technical, legal, and operational feasibility** of the proposed LDT solutions.
- **Validate stakeholder roles and governance structures** to ensure alignment and accountability.
- **Prioritise actions** based on feasibility, impact, and strategic alignment.
- **Set up a data governance and legal framework** for your LDT project.
- **Embed interoperability** by-design into the LDTs specifications.



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2. VALIDATE

Roles to involve

	Roles	Rationale
Necessary	**Project Coordination Roles** (Project Coordinators, etc.)	They are are vital to align stakeholders, manage timelines, and ensure seamless integration of validated outputs into the project's strategic roadmap.
	Domain Expertise Roles (Use-Case Owners, Subject-Matter Experts, Researchers, Field Agents, etc.)	They ensure the real-world relevance and accuracy of use cases, validating technical and operational assumptions against domain-specific knowledge.
	Data Roles (Geographic Information System Specialists, Data Engineers, Open Data Officers, Data Scientists, etc.)	They confirm data feasibility, quality, and interoperability, validating whether proposed data sources and processes meet project requirements.
	Technical Roles (Information Technology Department Staff, External Service Providers, Technical Experts, etc.)	They are able to assess feasibility, validate technical requirements, and ensure alignment between proposed solutions and operational realities.
	End Users (Citizens, Citizens' Associations or panels, etc.)	They provide user-centric validation, ensuring solutions align with actual needs, usability, and societal impact.
	Legal Roles (Data Protection Officer, Legal Experts, Intellectual Property Managers, etc.)	They are critical to ensure compliance with regulations, mitigate risks, and align data governance with legal and contractual obligations.

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2. VALIDATE

Workshops

#	Workshop	Duration
2.1	Co-Creating Effective Visualisation Dashboards	3h
2.2	Focus on Data Governance	*5 workshops (see following slide)*
2.3	Inventorying your LDT Project Data	1h 40min
2.4	Identifying Levers, Obstacles and Objectives for Interoperability	2h
BONUS	Lego® Serious Play	2h 45min

LDT4SSC Methodology

2. VALIDATE

Workshops

#	Workshop	Duration
2.2.1	Understanding Data Governance and Setting the Goal	2h
2.2.2	Mapping Stakeholders' Legitimacy and Authority	1h 30min
2.2.3	Designing your Data Governance Roadmap	2h 5min
2.2.4	Complementing your Data Governance Roadmap with a Legal Framework	2h 30min
2.2.5	Refining your Legal Framework for Data Governance	1h 30min – 2h

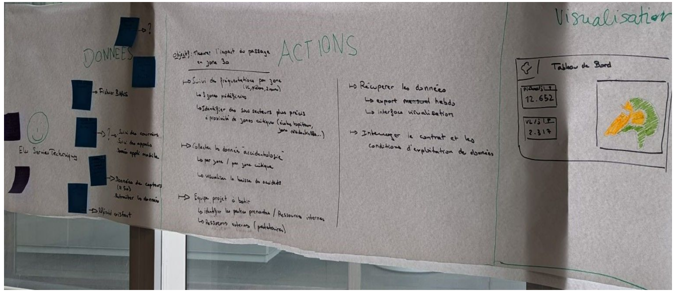
LDT4SSC Methodology

2. VALIDATE

Co-Creating Effective Visualisation Dashboards: Translating User Needs into Functional Requirements, Indicators and Visual Prototypes

Workshop Objectives

- 🎯 **Define end-user personas,** detailing their roles, constraints and decision-making contexts.
- 🎯 **Identify functional requirements** for the LDT based on end-user needs.
- 🎯 Translate the functional requirements into **indicators** and **visualisations** for dashboards.
- 🎯 Design dashboards and ensure they are **tailored** to different personas.



As part of the CAPACities programme, this workshop was held to identify the visualisations it wanted as part of its mobility use case. An 'elected official' persona and a 'technical services manager' persona were chosen, and the various maps were sketched out.

NB: Actual end users of the LDT can be invited to this workshop.

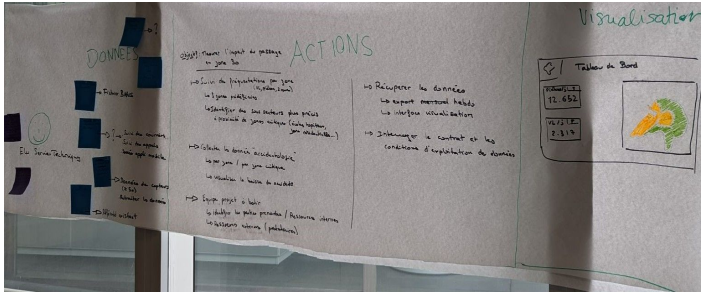
LDT4SSC Methodology

2. VALIDATE

Co-Creating Effective Visualisation Dashboards: Translating User Needs into Functional Requirements, Indicators and Visual Prototypes

Workshop Overview

Title	Co-Creating Effective Visualisation Dashboards: Translating User Needs into Functional Requirements, Indicators and Visual Prototypes
Methodological step	Step 2: VALIDATE (Specifications)
Duration	3 hours
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • Pens and markers (blue, green, red)
Expected outputs	• Identified **user personas** for the LDT. • **Functional requirements** for the LDT • **List of indicators** for the visualisations. • **Dashboards and visualisation mock-ups.** • **A summary note** produced after the workshop, outlining key findings and next steps.



As part of the CAPACities programme, this workshop identified the visualisations it wanted as part of its mobility use case. An 'elected official' persona and a 'technical services manager' persona were chosen, and the various maps sketched out.

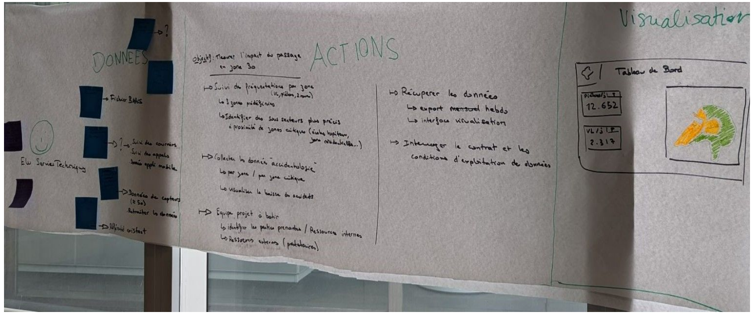
LDT4SSC Methodology

2. VALIDATE

Co-Creating Effective Visualisation Dashboards: Translating User Needs into Functional Requirements, Indicators and Visual Prototypes

Workshop Outline

Time spent	***Phase***	***Description***
15 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
20 mins	1. Identifying the personæs	Participants define one to three representative personas and their key characteristics.
30 mins	2. Identifying functional requirement	Participants list the key functional requirements for each persona.
30 mins	3. Choosing suitable Indicators and metrics	Participants identify potential indicators and metrics their persona would need to visualise.
1 hour	3. Designing Indicators and Visualising Dashboards	Participants design an initial mock-up of the dashboard.



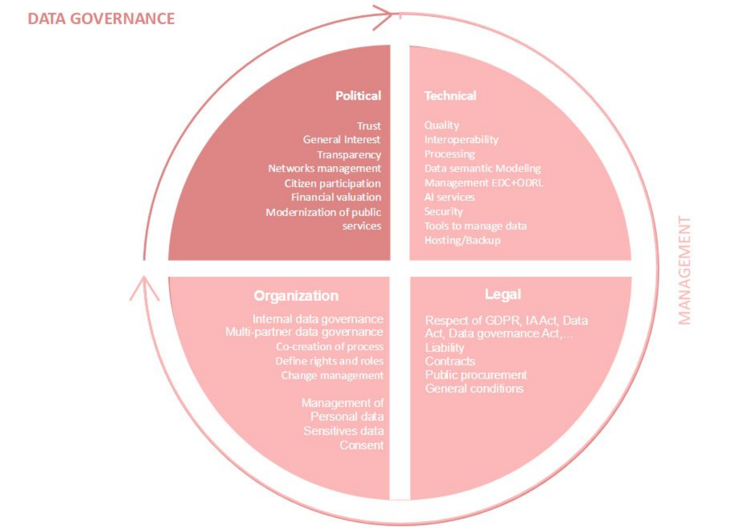
LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Understanding Data Governance and Setting the Goal: From Vision to a First Actionable Roadmap

Workshop Objectives

- 🎯 **Clarify the concept of data governance** by defining its key principles, components and significance within the consortium.
- 🎯 **Collectively** imagine and describe an ideal scenario where data governance is fully and successfully implemented.
- 🎯 **Determine the essential actions** required to achieve this ideal scenario.
- 🎯 **Identify potential obstacles** that could impede effective data governance.
- 🎯 **Develop practical solutions** to address these challenges.



During this workshop, the city projected themselves in 2030, when data is ‘a reality in the community’. Processes and a committee structures were put in place. The participants wondered how they had gotten there: a sponsor had provided the impetus, a strategy had been adopted, a recruitment policy had been launched, specific functions had been identified in the business and support functions.…

LDT4SSC

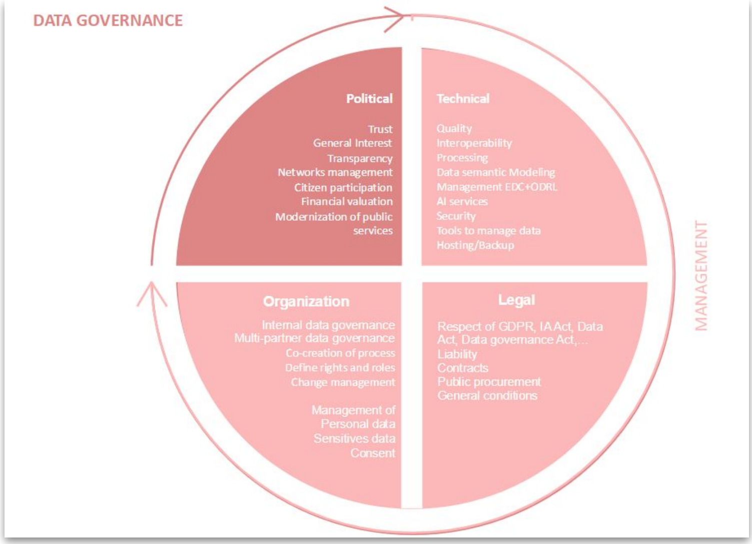
LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Understanding Data Governance and Setting the Goal: From Vision to a First Actionable Roadmap

Workshop Overview

Title	Understanding Data Governance and Setting the Goal: From Vision to a First Actionable Roadmap
Methodological step	Step 2: VALIDATE (Specifications)
Duration	2 hours
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • sticky-notes • Pens and markers
Expected outputs	• A **shared definition and scope of data governance** • A **future-state vision** (success and failure scenarios) • A **preliminary list of required actions** for a data governance strategy and roadmap • **A summary note** produced after the workshop, outlining key findings and next steps.



LDT4SSC

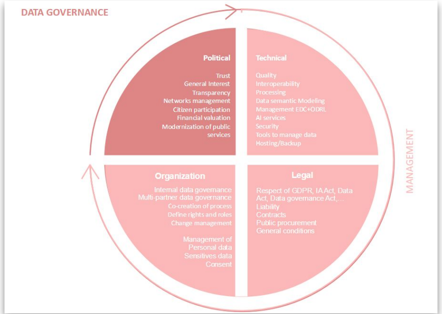
LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Understanding Data Governance and Setting the Goal: From Vision to a First Actionable Roadmap

Workshop Outline

Time spent	***Phase***	***Description***
10 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
30 mins	1. Brainstorming on Data Governance	Participants individually reflect on their understanding of data governance issues, while the facilitator collects, groups and summarises the ideas.
30 mins	2. Envisioning Different Scenarios	Participants envision themselves in five years: either the ideal fit-for-purpose data governance has been set up, or a failing one was set up.
30 mins	3. Understanding both the Actions taken to Achieve Success and the Missteps that Led to Failure	Participants map the critical stages —successes and pitfalls— over five years to shape actionable data governance strategies.
15 mins	4. Synthesising Outcomes and Defining Next Phases	Each subgroup presents its timeline and key insights, while the facilitator synthesises the discussion and defines follow-up actions.



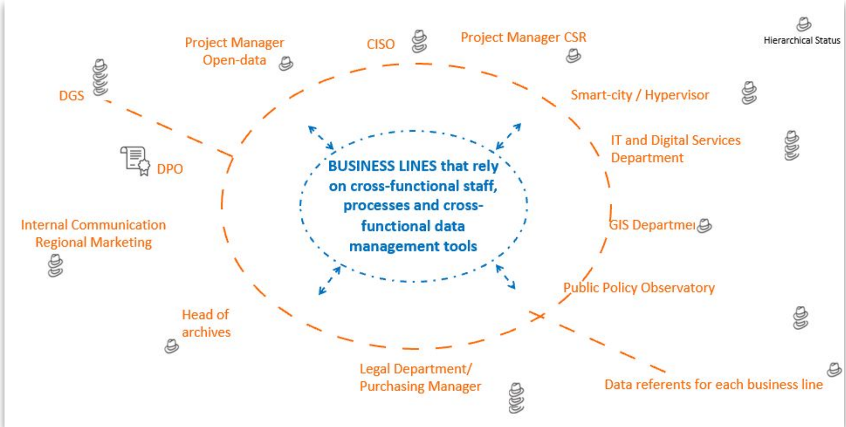
LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Mapping Stakeholders' Legitimacy and Authority to Act in Data Governance: Understand Roles, Responsibilities, and Hierarchical Influence

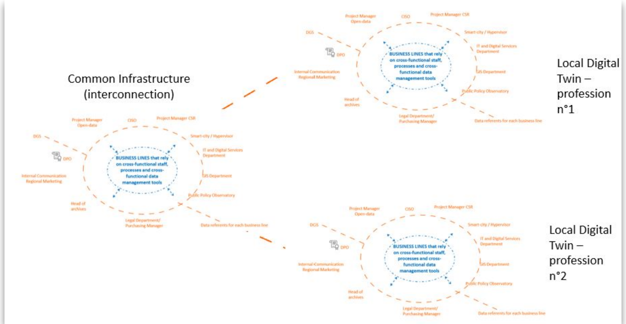
Workshop Objectives

- 🎯 Identify and **map stakeholders’ roles, responsibilities, and influence** in data governance.
- 🎯 Clarify **formal and informal authority** to ensure inclusive decision-making.
- 🎯 **Raise awareness** of differences in legitimacy and authority among **individuals working with data.**
- 🎯 **Map hierarchical positions** to facilitate the creation of a dedicated data governance unit and assign responsibilities.



Overview of the business lines to include in internal data governance

Data governance in LDT federations



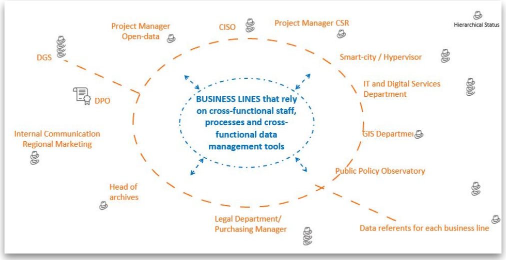
LDT4SSC Methodology

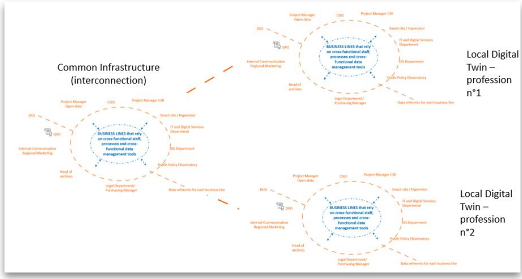
2. VALIDATE

DATA GOVERNANCE: Mapping Stakeholders' Legitimacy and Authority to Act in Data Governance: Understand Roles, Responsibilities, and Hierarchical Influence

Workshop Overview

Title	Mapping Stakeholders' Legitimacy and Authority to Act in Data Governance: Understand Roles, Responsibilities, and Hierarchical Influence
Methodological step	Step 2: VALIDATE (Specifications)
Duration	1 hour and 30 mins
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	● One big sheet of paper (1m/2m) or a large (digital) whiteboard ● Sticky notes ● Pens and markers
Expected outputs	● Data-related stakeholder map with legitimacy to act and hierarchical status ● List of key data governance participants. ● Preliminary assignment for data-governance roles ● **A summary note** produced after the workshop, outlining key findings and next steps





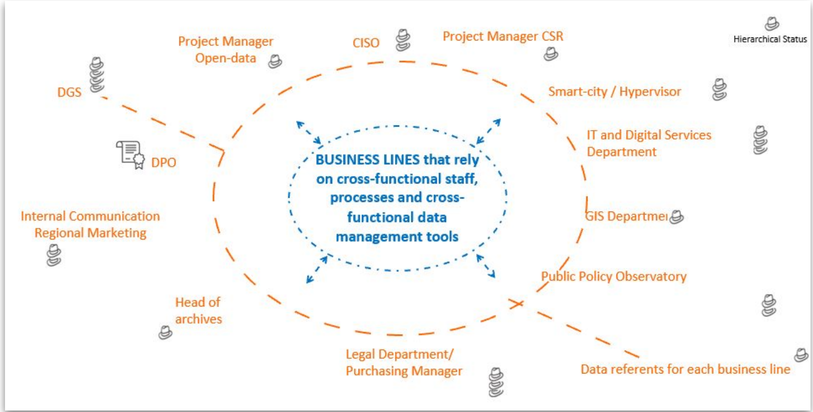
LDT4SSC Methodology

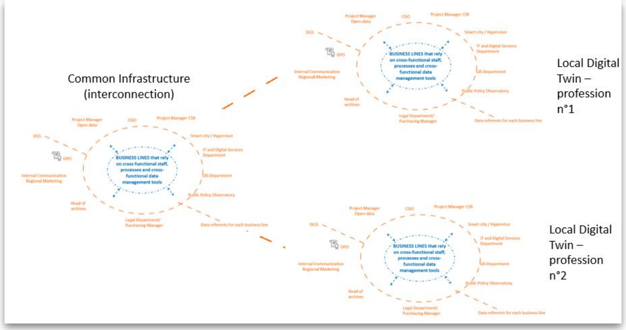
2. VALIDATE

DATA GOVERNANCE: Mapping Stakeholders' Legitimacy and Authority to Act in Data Governance: Understand Roles, Responsibilities, and Hierarchical Influence

Workshop Outline

Time spent	***Phase***	***Description***
15 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
30 mins	1. Mapping Contributors	Participants identify and describe the different roles involved in the data lifecycle and data governance
15 mins	2. Assigning Hats	Participants assign “hats” to individuals based on their hierarchical position to illustrate degree of decision-making authority
30 mins	3. Synthesising Insights and Assigning Governance Roles	The group summarises the workshops’ key findings, hence guiding the future development of data-governance and identifying the several individuals requiring formal mission statements.





LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Designing your Data Governance Roadmap: From Chief Data Officer Mission Statement to Action

Workshop Objectives

- 🎯 Define the role and **responsibilities of an ideal** (fictive) **Chief Data Officer**.
- 🎯 Develop a **detailed roadmap** using the Chief Data Officer's mission statement, covering **organisational, technical and human resource** aspects.
- 🎯 Assign **roles, timelines, and resources** for governance implementation.

from https://www.msicertified.com/blog/5w2h-training/

LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Designing your Data Governance Roadmap: From Chief Data Officer Mission Statement to Action

Workshop Overview

Title	Designing your Data Governance Roadmap: From Chief Data Officer Mission Statement to Action
Methodological step	Step 2: VALIDATE (Specifications)
Duration	2 hours and 5 mins
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	● One big sheet of paper (1m/2m) or a large (digital) whiteboard ● Sticky notes ● Pens and markers
Expected outputs	● Mission statement/job profile for the Chief Data Officer. ● Data-governance strategy roadmap with timeframes. ● List of roadmap actions for organisational (including HR), technical and legal aspects. ● **A summary note** produced after the workshop, outlining key findings and next steps.

from https://www.msicertified.com/blog/5w2h-training/

LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Designing your Data Governance Roadmap: From Chief Data Officer Mission Statement to Action

Workshop Outline

Time spent	***Phase***	***Description***
15 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
45 mins	1. Drafting the Mission Statement	Participants work in two subgroups to define the scope and functions of a data governance lead from different perspectives (internal authority vs. external partner).
45 mins	2. Build the roadmap	Participants map actions required to implement the mission statement across organisation, technical, and legal dimensions on a timeline
20 mins	3. Summarising Workshop findings and Defining next steps	The facilitator summarises the key findings and participants agree on the follow-up actions.

from https://www.msicertified.com/blog/5w2h-training/

LDT4SSC

LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Complementing your Data Governance Roadmap with a Legal Framework: From Mapping Legal Phases to Action

Workshop Objectives

- 🎯 **Understand the legal phases** of the LDT project.
- 🎯 **Align participants** on the legal framework for a successful project implementation.
- 🎯 Enrich the governance roadmap with **legal and regulatory requirements**.
- 🎯 Identify **areas that require further exploration** and discussion.

LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Complementing your Data Governance Roadmap with a Legal Framework: From Mapping Legal Phases to Action

Workshop Overview

Title	Complementing your Data Governance Roadmap with a Legal Framework: From Mapping Legal Phases to Action
Methodological step	Step 2: VALIDATE (Specifications)
Duration	2 hours and 30 mins
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • Sticky notes • Pens and markers
Expected outputs	• Visual map of project legal phases and associated actions • An initial legal framework integrated into the data-governance roadmap • **A summary note** produced after the workshop, outlining key findings and next steps.

LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Complementing your Data Governance Roadmap with a Legal Framework: From Mapping Legal Phases to Action

Workshop Outline

Time spent	***Phase***	***Description***
15 mins	0. Framing and Icebreaker	Participants are reminded of the chosen use case and the objectives of the project in order to anchor the reflection in a concrete context.
45 mins	1. Identifying Project Legal Phases	Participants map all legal stages of their project on a large circle on paper, while the facilitator may display technical stage diagrams for reference.
1 hour	2. Focusing on each Identified Stage	Participants analyse each legal stage, considering necessary clauses and contractual agreements, use circles to capture initial legal actions, and refer to a methodological guide for key questions.
30 mins	3. Synthesising Legal Insights and Defining Actionable Next Phases	Participants transform the workshop's legal insights into a structured roadmap, ensuring every phase of the project is compliant, actionable, and aligned with stakeholder responsibilities.

LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Refining your Legal Framework for Data Governance: In-Depth analysis of basic legal and contractual requirements

Workshop Objectives

- 🎯 Pinpoint the specific **legal and contractual requirements** for data governance.
- 🎯 Refine the legal framework to **mitigate risks and ensure compliance**.
- 🎯 Define **concrete, actionable courses of action**, translating identified needs into practical steps.
- 🎯 **Align these needs with existing legal clauses and available resources**, while identifying gaps that need to be addressed.
- 🎯 **Categorise each requirement** based on feasibility to create a **roadmap** for improving data management: **Can-be-resolved, Necessitate-Legislative-Changes, Demand-Collective-Action**.

LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Refining your Legal Framework for Data Governance: In Depth analysis of basic legal and contractual requirements

Workshop Overview

Title	Refining your Legal Framework for Data Governance: Deep dive into legal and contractual requirements
Methodological step	Step 2: VALIDATE (Specifications)
Duration	1 hour and 30 mins to 2 hours
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • A4 sheets • Pens and markers of several colors • Sticky notes
Expected outcomes	• A list of the projects’ legal requirements, challenges and issues. • An action plan for tackling these issues, requirements and challenges. • **A summary note** produced after the workshop, outlining key findings and next steps.

LDT4SSC Methodology

2. VALIDATE

DATA GOVERNANCE: Refining your Legal Framework for Data Governance: In Depth analysis of basic legal and contractual requirements

Workshop Outline

Time spent	***Phase***	***Description***
15 mins	0. Framing and Icebreaker	Participants are reminded of the chosen use case and the objectives of the project in order to anchor the reflection in a concrete context.
20 mins	1. Identifying Legal Requirements	Participants list their legal & regulatory requirements on post-it notes to identify missing contractual clauses, internal processes to be developed, etc.
30 mins	2. Developing a Structured Action Sheet	For each issue identified, participants complete an action sheet.
20 mins	3. Defining Next Phases	For each issue, participants identify possible next phases.
10 mins	4. Synthesising Outcomes and Wrapping Up	Each group presents its use case, the problems identified and the proposed solutions.

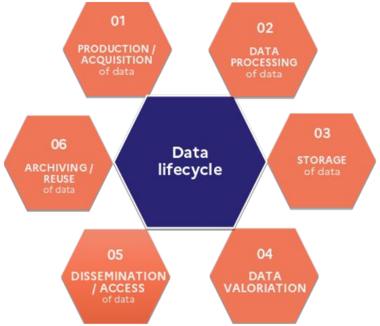
LDT4SSC Methodology

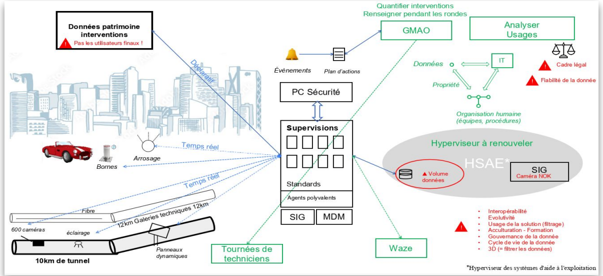
2. VALIDATE

Inventorying your LDT Project Data: Unlocking Interoperability, Listing and committing to share data among partners

Workshop Objectives

- 🎯 Identify **data sources, formats, and access protocols**.
- 🎯 Register **relevant data for the project** and **understand its lifecycle**.
- 🎯 Define and **formalize participant commitments** for **sharing data samples** with partners or service providers.
- 🎯 **Unlock interoperability** by listing and committing to share data among partners





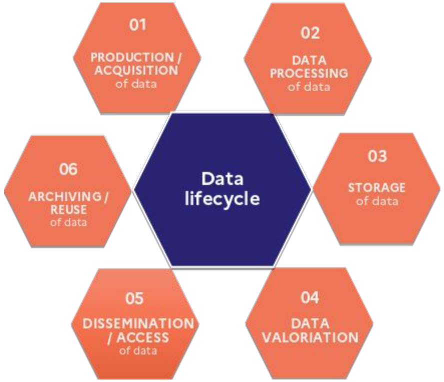
LDT4SSC Methodology

2. VALIDATE

Inventorying your LDT Project Data: Unlocking Interoperability, Listing and committing to share data among partners

Workshop Overview

Title	Inventorying your LDT Project Data: Unlocking Interoperability, Listing and committing to share data among partners
Methodological step	Step 2: VALIDATE (Specifications)
Duration	1 hour and 40 mins
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • Sticky notes • Pens and markers • paper
Expected outputs	• The filled-in data inventory table. • Workshop participant commitments for data sharing. • **A summary note** produced after the workshop, outlining key findings and next steps.



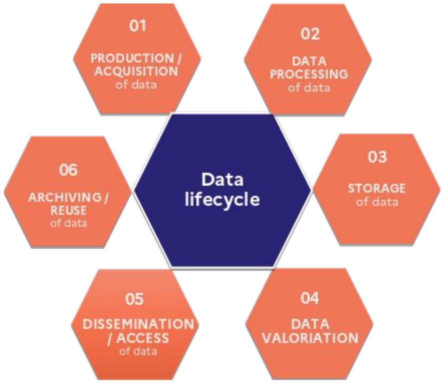
LDT4SSC Methodology

2. VALIDATE

Inventorying your LDT Project Data: Unlocking Interoperability, Listing and committing to share data among partners

Workshop Outline

Time spent	*Phase*	*Description*
15 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
45 mins	1. Inventorying the LDT Project's Data	Participants collectively fill in the data inventory table with datasets to be used in the LDT and their accompanying information
30 mins	2. Capturing Commitments through a Visual Thread	Participants write down their data-sharing commitments.
10 mins	3. Synthesising Outcomes and Defining Next Steps	The facilitator and participants review the findings of the workshop, summarise them and draft follow-up actions.

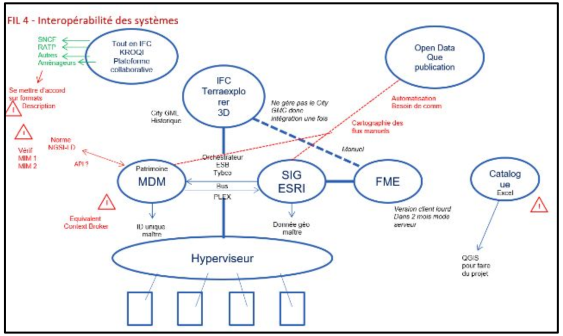


LDT4SSC Methodology

Identifying Levers, Obstacles and Objectives for Interoperability: From Awareness to Actionable Strategies for LDT Projects

Workshop Objectives

- 🎯 **Build a shared understanding of the information system's technical architecture**, ensuring all stakeholders have a clear view of its components and interactions.
- 🎯 **Identify the key improvements required to streamline data flows and interoperability** between systems, applications, and platforms, enabling more reliable and efficient information exchange.
- 🎯 **Lay the foundations for successful use-case implementation within the local government**, by ensuring the architecture can effectively support operational and business needs.



This workshop was carried out by 'Paris la Défense' as part of their *hypervisor overhaul* project. Their technical infrastructure was modelled, and the links between the different bricks formalised, identifying where there were problems. Participants were asked to identify the "ideal" building blocks they would need, which highlighted the need for a fairly central MDM-type infrastructure to manage the authority's reference data.

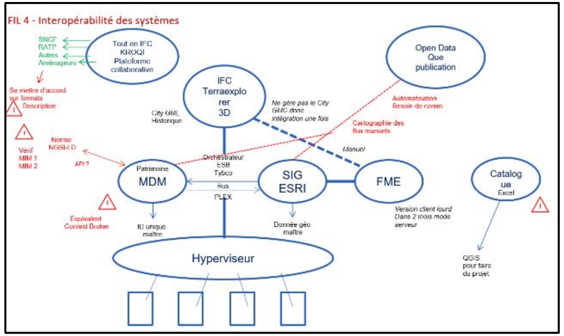
LDT4SSC Methodology

2. VALIDATE

Identifying Levers, Obstacles and Objectives for Interoperability: From Awareness to Actionable Strategies for LDT Projects

Workshop Overview

Title	Identifying Levers, Obstacles and Objectives for Interoperability: From Awareness to Actionable Strategies for LDT Projects
Methodological step	Step 2: VALIDATE (Specifications)
Duration	2 hours
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • Pens and colorful markers. • Sticky notes, • Voting stickers. • Potential digital tools (slides etc.).
Expected outputs	• **Photo/scan of the produced architecture diagram**. • List of concrete solutions to tackle technical interoperability issues. • **A summary note** produced after the workshop, outlining key findings and next steps.



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LDT4SSC

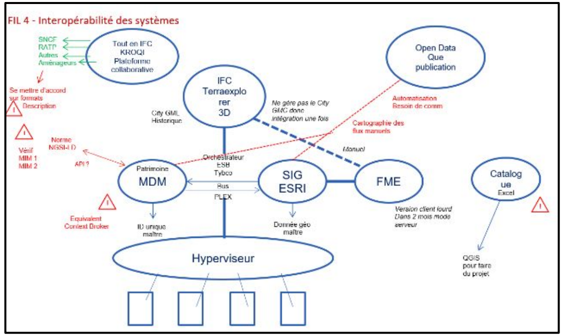
LDT4SSC Methodology

2. VALIDATE

Identifying Levers, Obstacles and Objectives for Interoperability: From Awareness to Actionable Strategies for LDT Projects

Workshop Outline

Time spent	*Phase*	*Description*
15 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
1 hour	1. Sketching the technical architecture	Participants sketch out the current information system architecture focusing on applications, software and data flow.
1 hour	2. Solving interoperability issues	Participants address issues identified in the previous phase of the workshop and come up with solutions.
15 mins	3. Synthesising and Defining Next Steps	The facilitator and participants review the findings of the workshop, summarise them and draft follow-up actions.



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LDT4SSC

LDT4SSC Methodology

2. VALIDATE

Bonus: Lego® Serious Play: A Transversal Tool to Address and Overcome Data Governance Challenges

Workshop Objectives

- 🎯 **Overcome complex challenges** and **deepen understanding** of data governance challenges.
- 🎯 Foster **creative problem-solving and stakeholder alignment**.
- 🎯 Identify **gaps, risks, and unresolved issues** before project implementation.
- 🎯 Co-create **concrete, user-oriented solutions** based on real challenges

LDT4SSC

LDT4SSC Methodology

2. VALIDATE

Bonus: Lego® Serious Play: A Transversal Tool to Address and Overcome Data Governance Challenges

Workshop Overview

Title	Lego® Serious Play: A Transversal Tool to Address and Overcome Data Governance Challenges
Methodological step	Step 2: VALIDATE (Specifications)
Duration	2 hours and 45 mins
Number of Facilitators	1 lead facilitator + 1 facilitator per group of 4 to 5 people
Equipment Needed	● Legos
Expected outputs	● Physical Lego models ● Recorded presentations, photos/videos for reuse. ● Summary note produced after the workshop, outlining key findings, solutions and next steps.

LDT4SSC Methodology

2. VALIDATE

Bonus: Lego® Serious Play: A Transversal Tool to Address and Overcome Data Governance Challenges

Workshop Outline

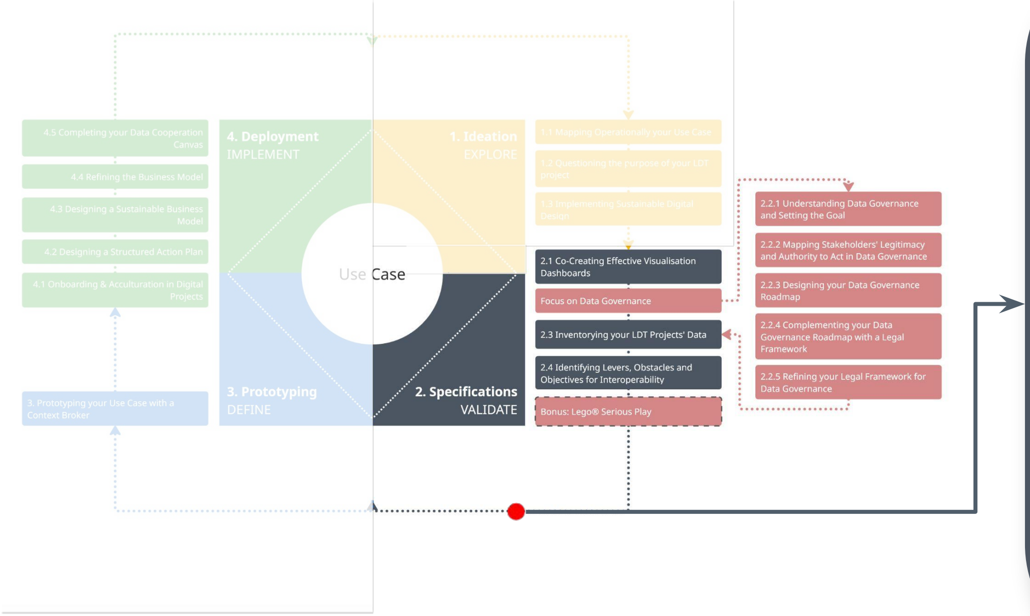
Time spent	***Phase***	***Description***
15 mins	0. Framing	Participants revisit the selected use case and project objectives to keep discussions grounded.
15 mins	1.Igniting Creativity through a Collaborative Lego Challenge	Participants start off with a short individual challenge aimed at (re)familiarising them with Legos, and then undertake a collective one to foster creativity and teamwork.
15 mins	2. Selecting Challenges and Forming Focused Working Groups	Participants, in subgroups, choose a LDT project-related challenge..
30 mins	3. Individual Prototyping of Solutions	Participants individually build their prototype solution to the challenge they chose using Lego bricks.
20 mins	4. Collectively Refining User-Centred Solutions	Participants regroup and debate on the best means and solutions to address their challenge, building on everyone's contributions from the previous phase.
30 mins	5. Presenting and Documenting Lego Models	Each subgroup presents the group their challenge, lego model, and the solutions they found to overcome it.
30 mins	6. Connecting Models	Participants connect all the lego models from the different sub-groups across challenges to broaden the perspectives.
10 mins	7. Synthesising Insights and Defining Next Steps	The facilitator summarises the key findings and participants agree on the follow-up actions.

LDT4SSC

LDT4SSC Methodology

2. VALIDATE

Achievements at the end of the VALIDATE step



At the end of this step of the methodology, the following elements have been completed or identified:

- **Technical, legal, and operational constraints** documented and addressed.
- **Refined stakeholder roles and governance structures** with clear responsibilities.
- **Actionable roadmaps** for data governance, and legal compliance.
- **Prioritised action plans** with timelines and resources for data governance and interoperability.

Note: At the end of this step, a specifications document/Terms of reference (ToR) can be drawn-up setting out the specs and requirements for the LDT.

3. DEFINE

The Prototyping step

LDT4SSC Methodology

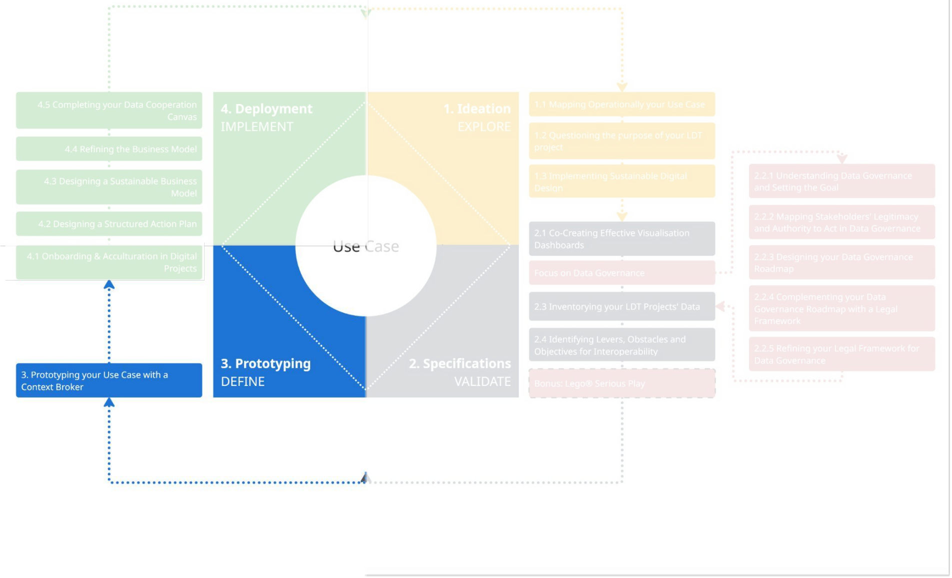
3. DEFINE

Introduction

Prototyping is the stage that transforms a **still theoretical project into concrete proof of feasibility**. It is not yet a question of producing a finished service, but of rapidly testing a **solution on a restricted perimeter**, using **real data**, confronting ideas with technical, functional and human reality.

This is not a compulsory stage. The previous stages can show that **solutions on the market do indeed meet the needs identified**, and that prototyping is not a required necessity. Nevertheless, an experimentation stage to test and validate the solutioncan always be useful before large-scale deployment.

In the CAPACities method, the aim of prototyping is twofold: to demonstrate that the **idea works**, and to check that it has **real value** for users. It's an **iterative, experimental and frugal approach**, which should enable you to learn quickly, without committing to heavy investment. The aim is also, from this stage onwards, to rely on a scalable infrastructure, so that you don't have to start from scratch during the deployment phase.

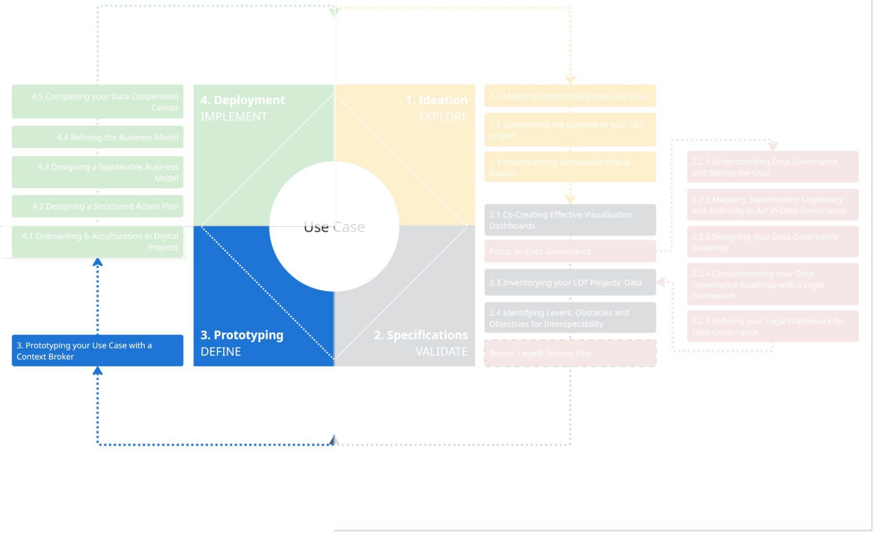


LDT4SSC Methodology

3. DEFINE

Prerequisites

- 📋 **Completed the VALIDATE step**.
- 📋 **Technical and operational expertise** to define and prototype system architecture, data flows, and integration requirements.
- 📋 **Sufficient resources for** prototyping and user-testing.

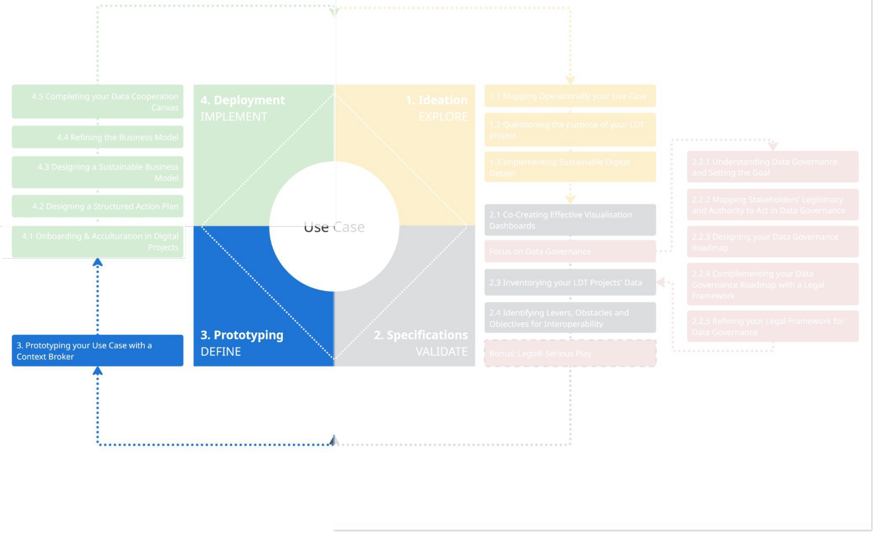


LDT4SSC Methodology

3. DEFINE

High Level objectives

- 🎯 **Formalise the LDT project's architecture** aligned with LDT interoperability Blueprint.
- 🎯 **Demonstrate that the LDT solution works** to answer the project's goals
- 🎯 **Check that the LDT solution has real values** for end users and answers their needs.
- 🎯 **Prepare for seamless transition** into the IMPLEMENT step with a fully documented and approved project blueprint.



LDT4SSC Methodology

3. DEFINE

Roles to involve

	Roles	Rationale
Necessary	**Project Coordination Roles** (Project Manager, etc.)	They Orchestrate cross-functional alignment, timelines, and resource allocation for definition outputs.
	Data Roles (GIS Specialists, Data Engineers, Open Data Officers, Data Scientists, etc.)	They design data architectures and workflows, validating interoperability and quality standards.
	Technical Roles (Information Technology Department Staff, External Service Providers, Technical Experts, etc.)	They build the technical architecture and solution and deploy the prototype and ensure best practices are followed.
	End Users (Citizens, Citizens' Associations or panels, etc.)	They validate user experience (UX), added value of the solution and accessibility of proposed dashboards or tools.
Advised	**Innovation / Smart City Roles** (Smart City Experts, UX/UI Designers, Innovation Laboratories, etc.)	They drive user-centric, innovative design and ensure solutions align with smart city standards and emerging best practices.
	Domain Expertise Roles (Use-Case Owners, Subject-Matter Experts, Researchers, Field Agents, etc.)	They refine use-case specifications and technical requirements, ensuring alignment with operational realities.

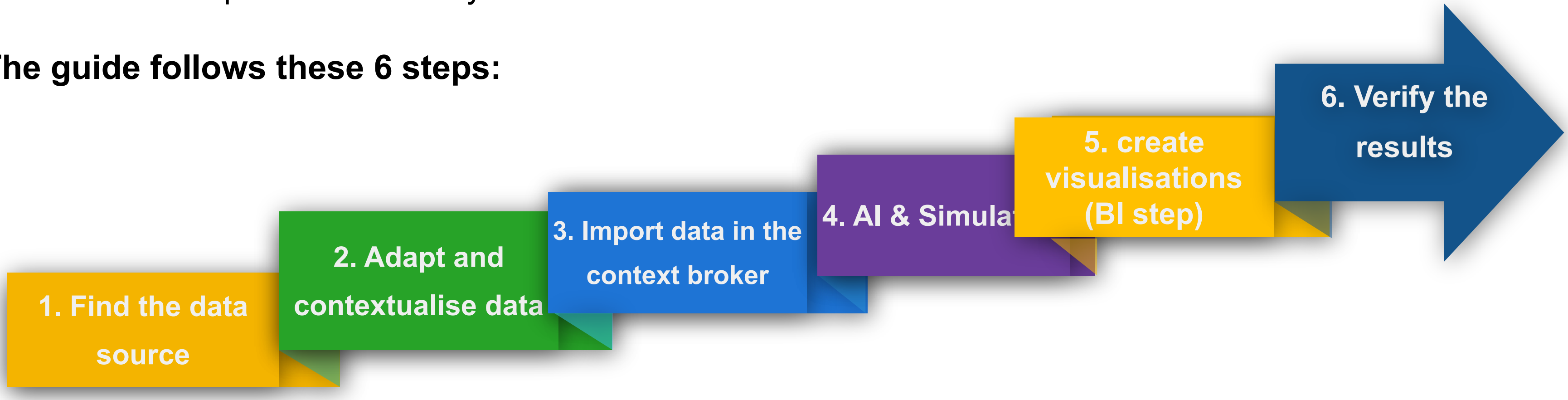
LDT4SSC Methodology

3. DEFINE

Example: Prototyping your Use Case with a Context Broker

A Step-by-Step Technical Guide based on a real use case was developed for LDT4SSC Pilots to illustrate this step. Not mandatory.

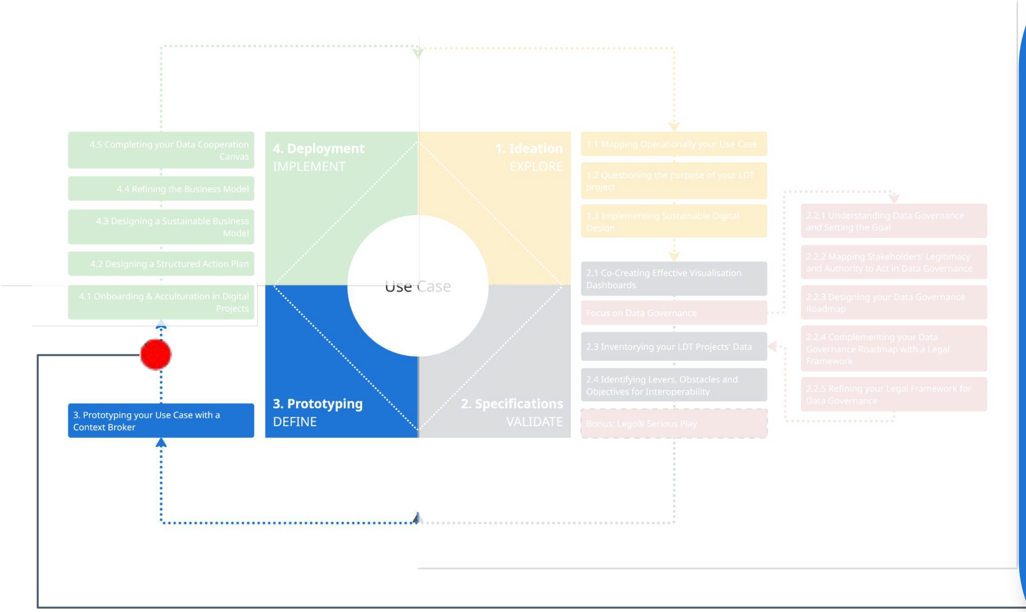
The guide follows these 6 steps:



LDT4SSC Methodology

3. DEFINE

Achievements at the end of the DEFINE step



At the end of this step of the methodology, the following elements have been completed or identified:

- A **prototype** that meets the needs of the end-users and objectives of the use case.
- Reusable and validated **technical and non-technical assets** to be used in deployment such as knowledge models, algorithms and data pipeline.
- **Documentation** of the prototype.

This step should end with **user testing**, in a logic of progressive test/iteration, to arrive at a validation from the community on scaling up.

LDT4SSC

4. IMPLEMENT

The Deployment step

LDT4SSC Methodology

4. IMPLEMENT

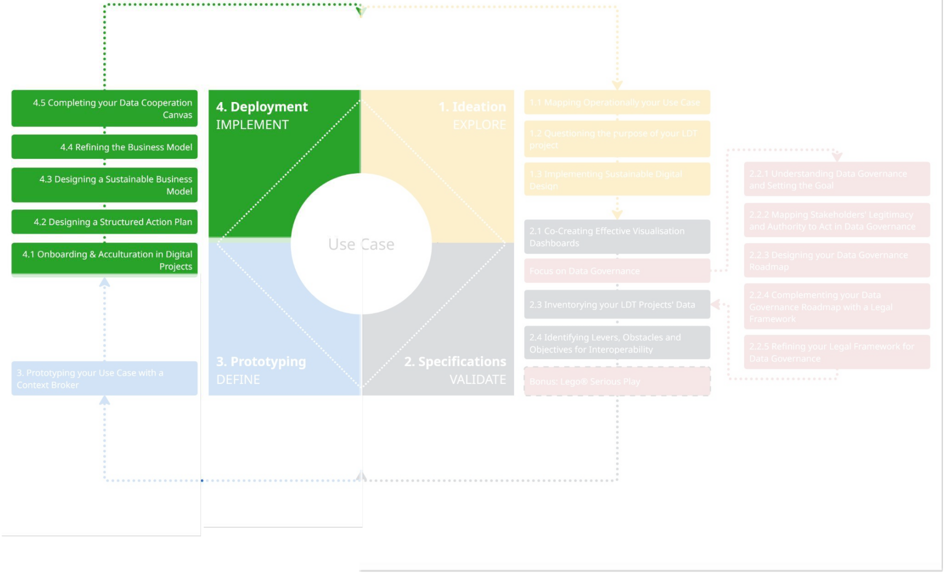
Introduction

Deployment is the final stage, marking the transition from experimentation to implementation. Having demonstrated the **feasibility** of a project through a prototype, it is now a question of getting it used by all the end users within the organisation. This phase involves **stabilising the prototype, making it reliable** and putting in place **maintenance** so that it becomes a long-term tool.

During this phase, the project's technical infrastructure, its **business model** and the legal framework for its use will be finalised, and a deployment **action plan** will be drawn up, involving in particular the adoption, training and onboarding of all staff.

Ideally, most of the subjects covered in this stage (business model, legal framework, staff familiarisation) **may already have been addressed upstream**, so the aim here is to formalise the concrete elements arising from these discussions.

This stage may also **enable the project to be scaled up**, either within the same region, or through re-use or pooling with other local authorities.

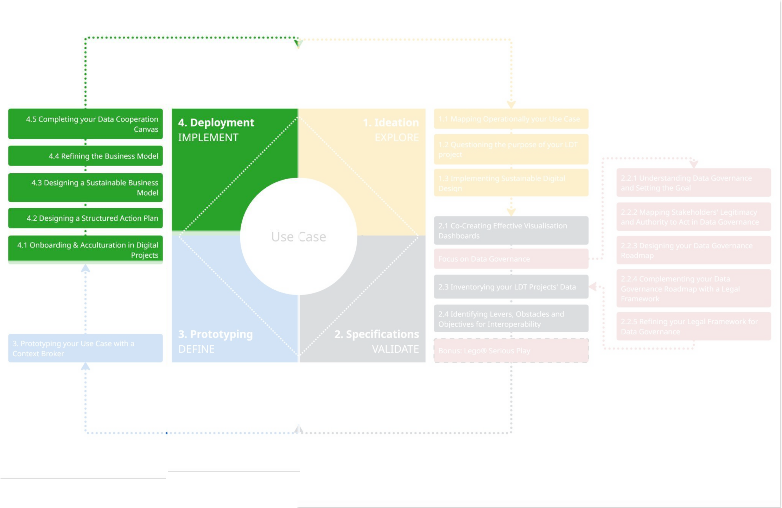


LDT4SSC Methodology

4. IMPLEMENT

Prerequisites

- 📋 **Completed the DEFINE step**.
- 📋 **Secured resources** (budget, technology, personnel) for deployment, operations and maintenance.
- 📋 **Stakeholder readiness and commitment**, including project teams, political involvement and clear communication channels.
- 📋 A **clear vision of the future governance** of the LDT solution.
- 📋 **Compliance with legal, ethical, and regulatory requirements**, including data protection and interoperability standards.

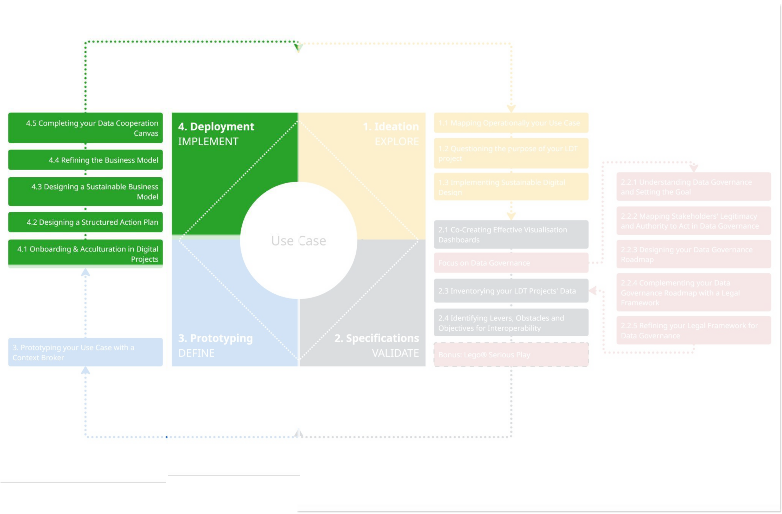


LDT4SSC Methodology

4. IMPLEMENT

High Level objectives

- 🎯 **Deploy the Local Digital Twin (LDT) platform** based on the defined architecture, governance frameworks, and action plans.
- 🎯 **Monitor and evaluate performance** against established KPIs, adjusting strategies as needed for optimal outcomes.
- 🎯 **Ensure stakeholder engagement and capacity-building** to support adoption, training, and long-term sustainability.
- 🎯 **Document lessons learned and best practices** for scalability, replication, and continuous improvement.
- 🎯 **Technically stabilise the solution** for use in production.
- 🎯 Organise the **maintenance, support and development** of the solution over time.
- 🎯 **Document** to encourage internal and external re-use.



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4. IMPLEMENT

Roles to involve

	Roles	Rationale
Necessary	**Project Coordination Roles** (Project Coordinators, etc.)	They oversee implementation timelines, stakeholder communication, and risk management.
	Data Roles (GIS Specialists, Data Engineers, Open Data Officers, Data Scientists, etc.)	They manage data pipelines, quality control, and real-time monitoring of the LDT system.
	Technical Roles (Information Technology Department Staff, External Service Providers, Technical Experts, etc.)	They execute platform deployment, integration, and troubleshooting, ensuring operational readiness.
	Legal Roles (Data Protection Officer, Legal Experts, Intellectual Property Managers, etc.)	They ensure ongoing compliance with data protection, licensing, and contractual obligations.
Advised	**Domain Expertise Roles** (Use-Case Owners, Subject-Matter Experts, Researchers, Field Agents, etc.)	They support field-level validation and troubleshooting of use-case-specific issues.
	End Users (Citizens, Citizens' Associations or panels, etc.)	They provide feedback on usability and impact, enabling iterative improvements post-deployment.
	Other Local Authorities facing similar issues	They enable cross-city knowledge exchange and replication of successful deployments, fostering scalability and shared learning.

LDT4SSC Methodology

4. IMPLEMENT

Workshops

#	Workshop	Duration
4.1	Onboarding & Adoption in Digital Projects	1h 30min
4.2	Designing a Structured Action Plan	1h 45min
4.3	Designing a Sustainable Business Model	1h 50min
4.4	Refining the Business Model	2h 25min

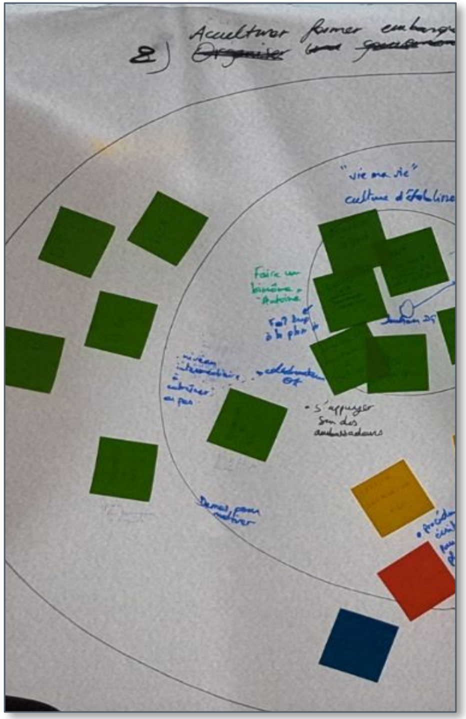
LDT4SSC Methodology

4. IMPLEMENT

Onboarding & Adoption in Digital Projects: Engaging Stakeholders, Defining Training Paths, and Building a Sustainable Adoption Plan

Workshop Objectives

- 🎯 **Identify the different target groups** that need to be made aware, trained, or mentored.
- 🎯 **Define the specific needs of each target group** in terms of knowledge and tools.
- 🎯 Explore **concrete and inspiring solutions** to foster buy-in and skill development.
- 🎯 Collaboratively develop an **action plan tailored to the different target groups** to effectively build awareness and provide training.
- 🎯 Develop **operational solutions** that can be directly implemented in a local city or community.



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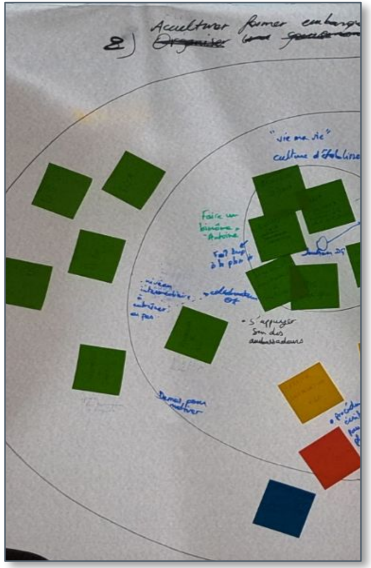
LDT4SSC Methodology

4. IMPLEMENT

Onboarding & Adoption in Digital Projects: Engaging Stakeholders, Defining Training Paths, and Building a Sustainable Adoption Plan

Workshop Overview

Title	Onboarding & Adoption in Digital Projects: Engaging Stakeholders, Defining Training Paths, and Building a Sustainable Adoption Plan
Methodological step	Step 4: IMPLEMENT (Deployment)
Duration	1 hour and 30 mins.
Number of Facilitators	2 facilitators per group
Equipment Needed	● One big sheet of paper (1m/2m) or a large (digital) whiteboard ● Markers and pens (multiple colors) ● Colored sticky notes
Expected outputs	● A structured action plan for training and onboarding of project stakeholders. ● Workshop summary note consolidating all outputs for implementation and follow-up: ○ Stakeholder involved in the project ○ Needs analysis for each stakeholder group ○ Concrete solutions and engagement levers identified for each group.



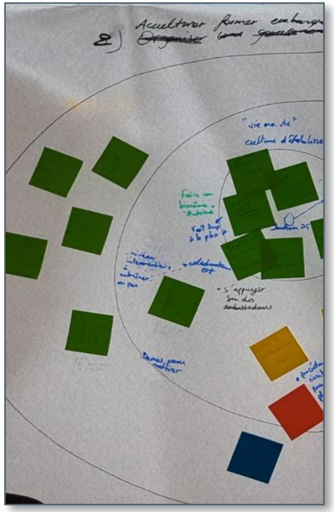
LDT4SSC Methodology

4. IMPLEMENT

Onboarding & Adoption in Digital Projects: Engaging Stakeholders, Defining Training Paths, and Building a Sustainable Adoption Plan

Workshop Outline

Time spent	***Phase***	***Description***
15 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
15 mins	1. Identifying Targets	Participants identify three different levels of stakeholders involved in the project.
15 mins	2. Defining Needs	Participants define the identified stakeholders needs.
20 mins	3. Identifying Solutions and Levers	Participants identify concrete actions, tools and methods to address the aforementioned needs.
20 mins	4. Drafting a Finalised Action Plan	Participants build on the solutions and levers previously identified to collectively develop an adoption and training plan per target stakeholder group.
10 mins	5. Synthetising Insights and Defining Next Steps	The facilitator summarises the key findings and participants agree on the follow-up actions.



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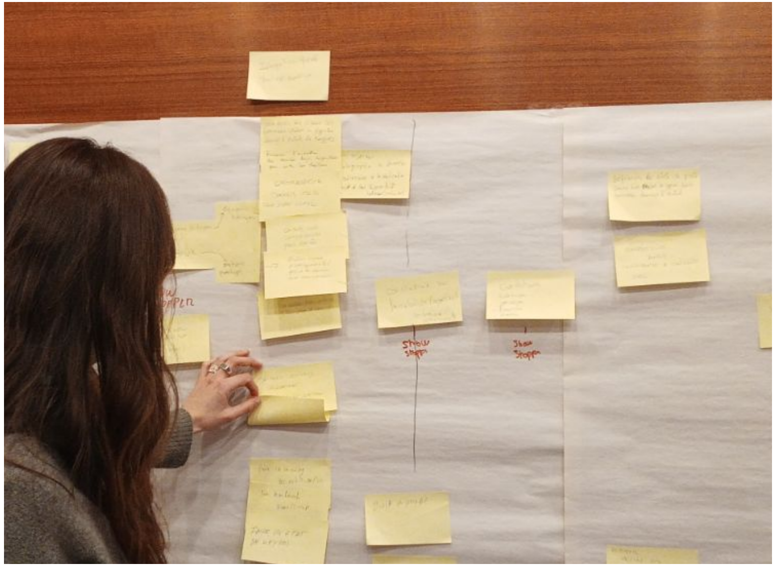
LDT4SSC Methodology

4. IMPLEMENT

Designing a Structured Action Plan: From Ideas to Execution for Your Use Case

Workshop Objectives

- 🎯 Collectively identify the actions (legal, technical, organisational) needed to implement the LDT project.
- 🎯 Organize these steps into a realistic timeline (short-, medium-, or long-term).
- 🎯 Develop various implementation scenarios tailored to the constraints and opportunities.
- 🎯 Lay a solid foundation for a report or presentation intended for elected officials and decision-makers.



The workshops helped to highlight the chronology of actions to be taken, from the most immediate to the most distant, as well as the various milestones to be put in place. They also helped to sort out what was desirable from what was achievable, prioritizing the actions that were truly essential.

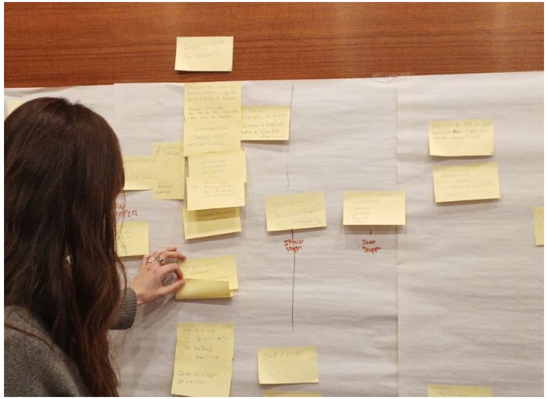
LDT4SSC Methodology

4. IMPLEMENT

Designing a Structured Action Plan: From Ideas to Execution for Your Use Case

Workshop Overview

Title	Designing a Structured Action Plan: From Ideas to Execution for Your Use Case
Methodological step	Step 4: IMPLEMENT (Deployment)
Duration	1 hour and 45 mins
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • A4 sheets • Pens and markers
Expected outputs	• A Photo/scan of the produced timeline diagram. • An action plan with 3 potential implementation scenarios • A **clean, digital version** of the timeline diagram (done by the facilitator after the workshop). • **A summary note** produced after the workshop, outlining key findings and next steps.



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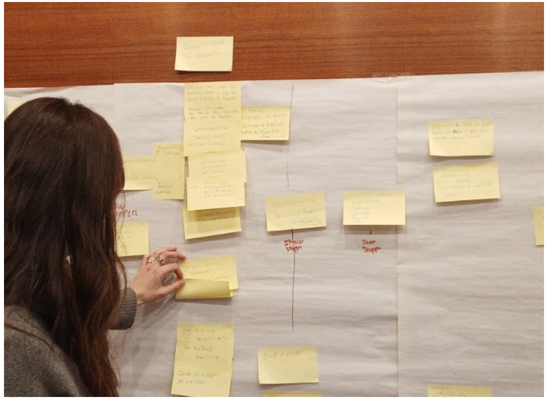
LDT4SSC Methodology

4. IMPLEMENT

Designing a Structured Action Plan: From Ideas to Execution for Your Use Case

Workshop Outline

Time spent	***Phase***	***Description***
15 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
15 mins	1. Identifying Required Actions	Participants individually reflect on all the necessary actions needed in order to implement the project.
30 mins	2. Structuring the Action Plan	Participants organise a visual chronological plan with the actions to be done.
30 mins	3. Developing Implementation Scenarios	Participants develop three implementation scenarios in order to be able to present alternative options that can be submitted to decision-makers for arbitration.
15 mins	4. Synthetising Findings and Defining Next Steps	The facilitator summarises the key findings and participants agree on the follow-up actions.



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LDT4SSC Methodology

4. IMPLEMENT

Designing a Sustainable Business Model: Funding and Valuation Strategies for Your Project

Workshop Objectives

- 🎯 Help stakeholders identify and **structure the financial levers** that will support the sustainability of their project.
- 🎯 Develop a **funding and valuation strategy** for the project.

This workshop, held during CAPACities 2, identified all sources of project costs and available sources of funding: own budget, pooling of resources, public-private partnerships, external grants, savings generated, sale of services to third parties, etc.

LDT4SSC Methodology

4. IMPLEMENT

Designing a Sustainable Business Model: Funding and Valuation Strategies for Your Project

Workshop Overview

Title	Designing a Sustainable Business Model: Funding, and Valuation Strategies for Your Project
Methodological step	Step 4: IMPLEMENT (Deployment)
Duration	1 hour and 50 minutes
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • Sticky notes • Pens and markers • Stickers
Expected outputs	• List of Identified and prioritised Funding Opportunities • Inputs on data and service value for the LDT's business model • Summary note produced after the workshop, outlining key findings and next steps that will help build the business plan

This workshop, held during CAPACities 2, identified all sources of project costs and available sources of funding: own budget, pooling of resources, public-private partnerships, external grants, savings generated, sale of services to third parties, etc.

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LDT4SSC Methodology

4. IMPLEMENT

Designing a Sustainable Business Model: Funding and Valuation Strategies for Your Project

Workshop Outline

Time spent	***Phase***	***Description***
15 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
20 mins	1. Identifying and Categorising Funding Opportunities	Participants reflect on potential funding sources for sustaining the LDT.
30 mins	2. Assessing Data and Service Business Value	Participants evaluate the potential value of the LDTs data and services.
30 mins	3. Prioritising funding and valuation strategies	Participants collectively identify the most promising and sustainable solutions for project viability.
15 mins	4. Synthesising Findings and Defining Next Steps	Participants summarise the key findings and draft follow-up actions.

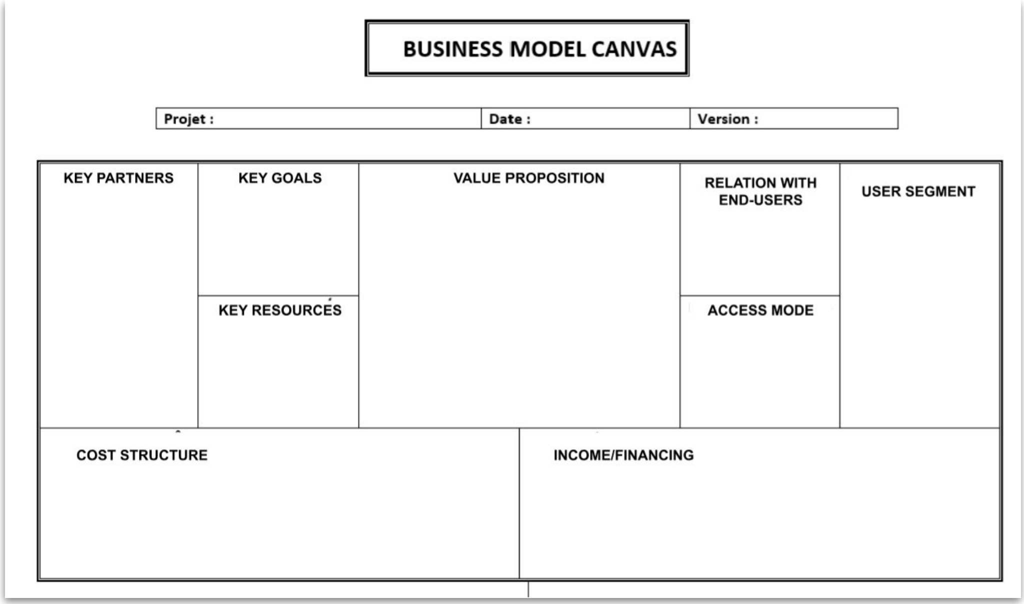
LDT4SSC Methodology

4. IMPLEMENT

Refining the Business Model: Designing, Testing, and Refining Your Strategy with a Canvas

Workshop Objectives

- 🎯 **Fill-in and refine** the **Business Model Canvas** as a mean to present the project's business model
- 🎯 Align the model with the project's **strategic goals and stakeholder needs**.



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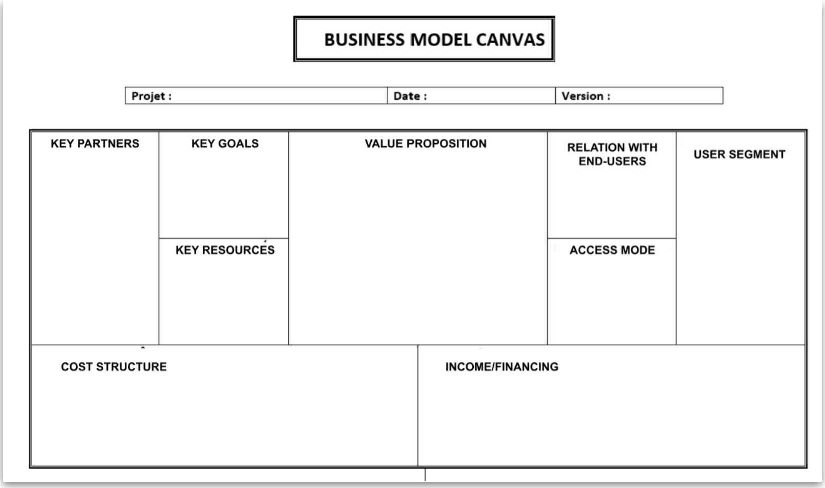
4. IMPLEMENT

Refining the Business Model: Designing, Testing, and Refining Your Strategy with a Canvas

Workshop Overview

Title	Refining the Business Model: Designing, Testing, and Refining Your Strategy with a Canvas
Methodological step	Step 4: IMPLEMENT (Deployment)
Duration	2 hours and 25 mins
Number of Facilitators	1 for each group of 4 to 5 people
Equipment Needed	• One big sheet of paper (1m/2m) or a large (digital) whiteboard • Sticky notes • Pens and markers • paper
Expected outputs	• Completed Business Model Canvas • **A summary note** produced after the workshop, outlining key findings and next steps.



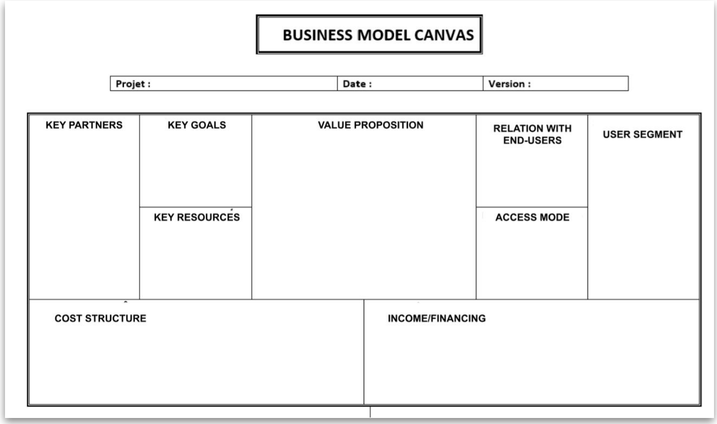
LDT4SSC Methodology

4. IMPLEMENT

Refining the Business Model: Designing, Testing, and Refining Your Strategy with a Canvas

Workshop Outline

Time spent	*Phase*	*Description*
15 mins	0. Framing and Icebreaker	Participants revisit the selected use case and project objectives to keep discussions grounded.
15 mins	1. Understanding the Business Model Canvas	The facilitator introduces the nine blocks of the Business Model Canvas so as to ensure every participant's understanding.
45 mins	2. Populating the Business Model Canvas	Participants brainstorm and populate each block with relevant information.
30 mins	3. Reviewing and Iterating the Business Model Canvas	Participants revisit each block, refine assumptions, and adjust them based on feedback or new insights.
15 mins	4. Synthesising Outcomes and Defining Next Steps	Participants summarise the key findings and draft follow-up actions.



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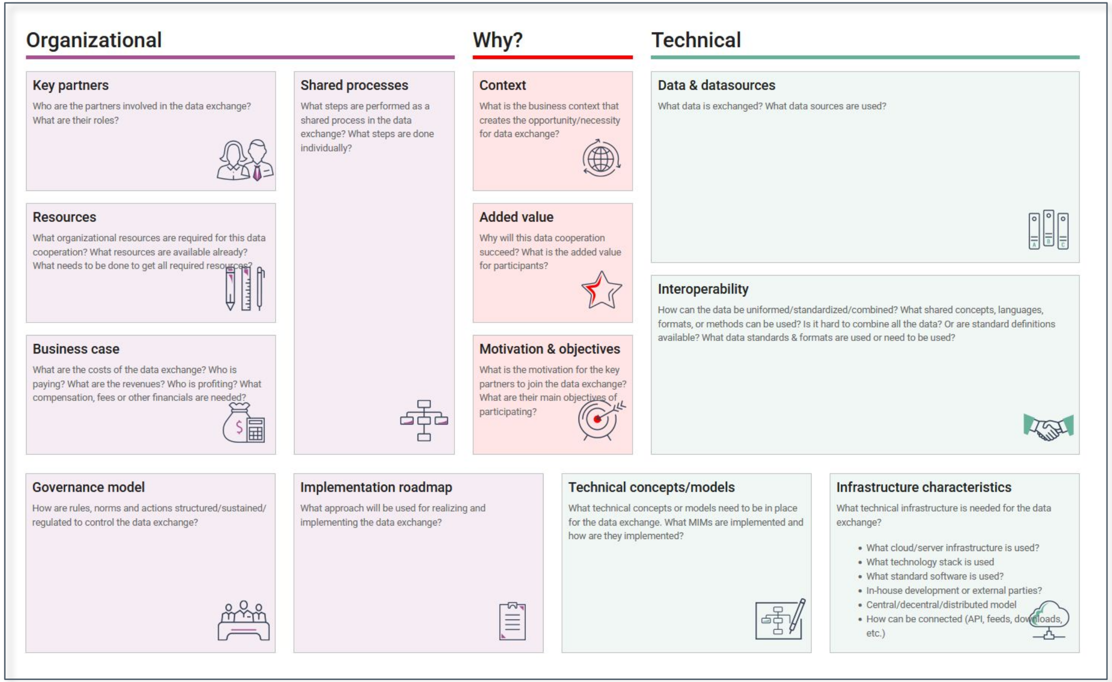
4. IMPLEMENT

Completing your Data Cooperation Canvas (DS4SSCC): Developing Multi-Stakeholder Data Cooperation

Objectives

- Define **purpose, scope, and value** of the data cooperation initiative.
- Structure the initiative using the **Data Cooperation Canvas** (Why, Who, What, How).

There is no workshop brief, you can find all the information needed here: https://www.datacooperationcanvas.eu/



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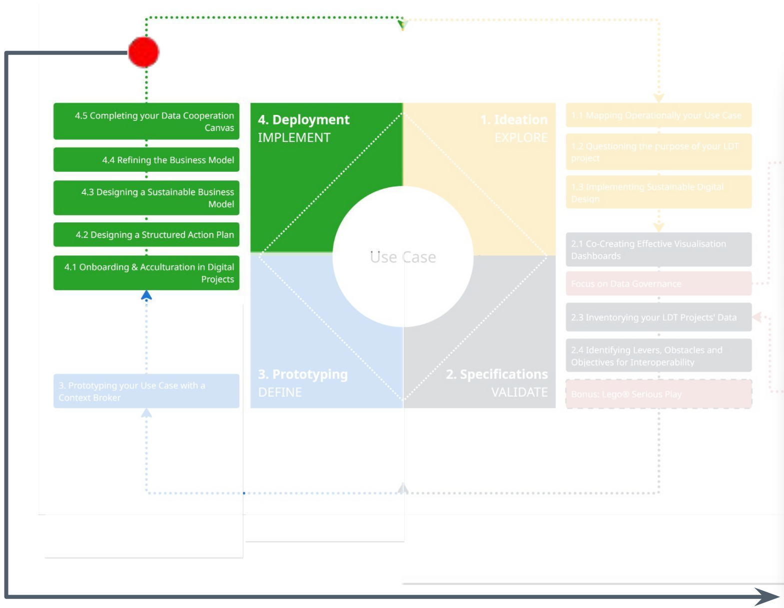
4. IMPLEMENT

Achievements at the end of the IMPLEMENT step



At the end of this step of the methodology, the following elements have been achieved or identified:

- An **action plan to scale up** the project (onboarding, adoption, next steps, links to other use cases, etc.).
- A **business model** to ensure the **project's long-term viability** and the foundations of their business plan.
- A **Deployed LDT** with integrated data sources and operational workflows ensuring quality, interoperability, and real-time monitoring.
- **Functional dashboards and tools** tailored to end user needs.
- **Trained stakeholders** (staff, partners, end-users) with capacity-building initiatives.

Note: during this step, completing the Data Cooperation Canvas (DS4SSCC) for establishing Multi-Stakeholder Data Cooperation should be done.

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Conclusion

The **LDT4SSC Methodology** provides a**, use case-centered framework** to design, validate, and deploy **Local Digital Twins** that address real-world local challenges.

By structuring the process into **four iterative steps —1-EXPLORE, 2-VALIDATE, 3-DEFINE, 4-IMPLEMENT**— it ensures:

✅ **Interoperable & replicable by design:** assets you build can be shared with other city and communities.

✅ **Concrete outputs every step:** you finish each step and workshop with a deliverable.

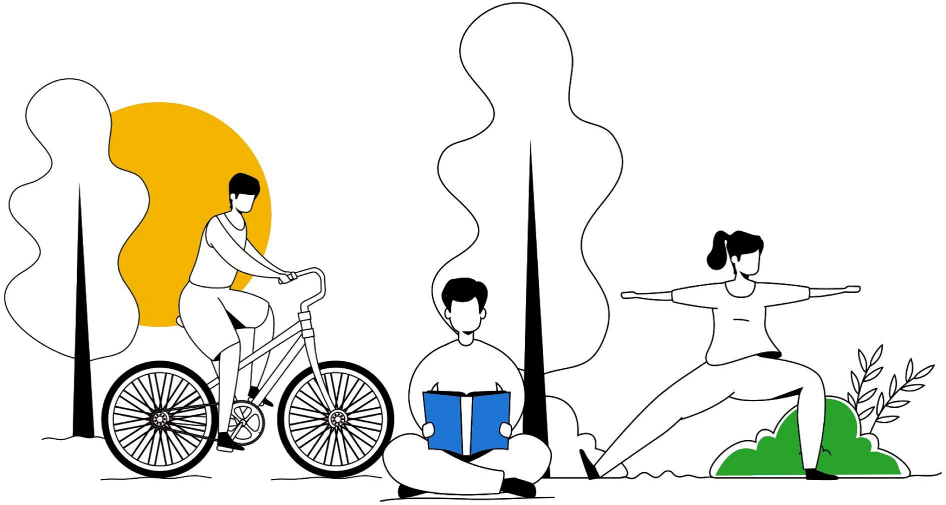
✅ **Flexible to your maturity:** run the full path or just the workshops you need. The path is non-linear, you can iterate and loop back.

✅ **Aligns everyone:** business, technical, legal, decision-makers, citizens, in the same room.

LDT4SSC Methodology

Please share your feedback !

Please **share your feedback on the Methodology** and its materials to the LDT4SSC consortium **to help us improve it** for reuse by other cities and communities.



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Local Digital Twins for Smart Sustainable Communities

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